

Evaluation of Zearn Supplemental with Dedicated Implementation Support

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EXECUTIVE SUMMARY

In 2024, the Center for Research and Reform in Education (CRRE) at Johns Hopkins University worked with the Louisiana Department of Education (LDOE) and Zearn to conduct a quasi-experimental study of the Zearn Math Supplemental with dedicated implementation support platform in Grades 4-8 funded by Accelerate.

Zearn is a nonprofit, online mathematics platform used by roughly 25% of elementary school students and more than one million middle-school students nationwide. Teachers using Zearn Math assign students lessons to complete on the platform. Lessons cover every concept of K-8 math. Upon logging in, students engage in guided lesson activities designed to enhance their understanding of mathematical concepts and procedures. All Zearn Math lessons include fluency games, videos led by on-screen teachers with pause points to solve mathematics problems, and a closing mastery-based quiz on which students must score 100% to advance to the next lesson. Students are prompted to solve problems both on screen and on paper. If students get stuck, the lessons adapt in real time to give students the support they need.

Zearn Math is primarily used as a supplemental digital program for mathematics instruction, combining digital lessons with printed student materials. Students typically complete Zearn digital lessons during designated instructional time such as the mathematics block, intervention sessions, tutoring blocks, or other designated supplemental learning times.

In some implementations, Zearn Math is paired with dedicated implementation support, which includes professional development and training for teachers and school leaders, as well as school-level coaching (“Zearn Supplemental”). This support helps establish school-wide routines for scheduling, data analysis, and student engagement, all designed to help students complete three grade-level digital lessons per week—Zearn’s recommended usage. A recent study, *Implementation of Zearn Math for Supplemental Use in Louisiana* (Ross et al., 2024), examined how the program was implemented in three Louisiana districts and explored stakeholder perceptions of both Zearn Math and the accompanying implementation supports.

The focus of this evaluation was three Louisiana school districts in which this supplemental program with dedicated implementation support was formally piloted: Lincoln Parish Schools, St. John the Baptist Parish Schools, and Natchitoches Parish Schools. These districts and the students and schools within the districts served as the treatment group in this study. The study examined research questions related to the impact of Zearn Supplemental in improving student mathematics achievement, both overall and for student subgroups, as well as how usage of Zearn Supplemental impacted student learning. In the following sections, we present an overview of the findings from this evaluation, which include results drawn from student academic achievement and program usage data.

- The present study employed a quasi-experimental design with propensity score



matching to evaluate the impact of Zearn Supplemental. Multi-level matching was conducted in two rounds. First, schools within districts were matched based on prior achievement and demographic variables to form three sets of district comparison groups. This allowed for the comparison of Zearn Supplemental users with non-Zearn users (Comparison Group 1), Zearn Supplemental users with Zearn users without implementation support (Comparison Group 2), and Zearn Supplemental users with “intended” Zearn users (Comparison Group 3).

- Comparison Group 1, comparing students in districts using Zearn Supplemental with non-Zearn using districts, defined non-user districts as districts with fewer than one lesson completed per week in prior Zearn usage. Comparison Group 2, comparing Zearn Supplemental districts with Zearn districts without implementation support that had one or more lessons completed per week on average. Finally, Zearn identified a set of districts that intended to fully implement Zearn district wide. These districts were used in the creation of Comparison Group 3.
- In the second stage of matching to create the analytic samples for each comparison group created through the first round of matching, students within schools were matched across districts based on student and school factors from the first round of matching.
- In the 2023-24 school year, the three treatment districts implemented the Zearn Supplemental with a dedicated implementation support platform for Grades K-8. The program model included school-level coaching to promote and guide high-fidelity tutoring usage.
- Data sources for this report included student achievement data, Zearn usage data, and student and school demographic data.
- Quantitative impact analyses showed that statistically significant, positive Zearn Supplemental program impacts were evidenced compared to non-Zearn user students (Comparison Group 1) and Zearn user students without implementation support (Comparison Group 2) in Grades 4-8. A positive but not statistically significant impact was seen in Comparison Group 3 between Zearn Supplemental users and Zearn-identified intended users.
- We conducted an exploratory analysis examining the association of the level of Zearn usage with achievement regardless of the treatment condition students, in which adequate usage was defined as completing 90 or more on-grade-level lessons. Analysis found that completing 90 or more lessons was associated with higher mathematics achievement at endline.
- Further exploratory analyses related to usage, again regardless of treatment condition, broke completed Zearn on-grade-level lessons down into four categories: Negligible (fewer than 30 lessons completed), Low Usage (between 30 and 59 lessons completed), Medium Usage (60-89 lessons completed), and High Usage (90 or more lessons completed).
- Analyses suggested that higher average lessons completed per week were also associated with higher mathematics achievement at endline. In addition, in all three comparison groups, completing 60 or more lessons was also associated with higher mathematics achievement at endline compared to students who completed fewer than 30 lessons.



INTRODUCTION

Overview of Zearn Math Supplemental

Zearn is a nonprofit education organization that created Zearn Math, an online mathematics platform used by roughly 25% of elementary school students and more than one million middle-school students nationwide. Teachers using Zearn Math assign students lessons to complete on the platform. Lessons cover every concept of K-8 math. Upon logging in, students engage in guided lesson activities designed to enhance their understanding of mathematical concepts and procedures. All Zearn Math lessons include fluency games, videos led by on-screen teachers with pause points to solve mathematics problems, and a closing mastery-based quiz on which students must score 100% to advance to the next lesson. Students are prompted to solve problems both on screen and on paper. If students get stuck, the lessons adapt in real time to give students the support they need.

Zearn Math is primarily used as a supplemental digital program for mathematics instruction, combining digital lessons with printed student materials. Students typically complete Zearn digital lessons during designated instructional time such as the mathematics block, intervention sessions, tutoring blocks, or other designated supplemental learning times.

In some implementations, Zearn Math is paired with dedicated implementation support, which includes professional development and training for teachers and school leaders, as well as school-level coaching. This support helps establish school-wide routines for scheduling, data analysis, and student engagement, all designed to help students complete three grade-level digital lessons per week—Zearn’s recommended usage. A recent study, *Implementation of Zearn Math for Supplemental Use in Louisiana* (Ross et al., 2024), examined how the program was implemented in three Louisiana districts and explored stakeholder perceptions of both Zearn Math and the accompanying implementation supports.

Overview of the Evaluation

The Center for Research and Reform in Education (CRRE) at Johns Hopkins University worked with the Louisiana Department of Education (LDOE) and Zearn to conduct a quasi-experimental study of Zearn Supplemental with dedicated implementation support platform in Grades K-8 funded by Accelerate. The focus was three school districts in which this supplemental implementation support program was being formally piloted: Lincoln Parish Schools, St. John the Baptist Parish Schools, and Natchitoches Parish Schools. The program model examined included school-level coaching to promote and guide high-fidelity tutoring usage. The achievement measure was the Louisiana Educational Assessment Program (LEAP) mathematics assessment conducted by the State of Louisiana for Grades 3-8.

The present study used a quasi-experimental propensity score matching (PSM) design with hierarchical linear models (HLM) analysis to examine the following research



questions:

1. What is the impact of using Zearn Supplemental with dedicated implementation support on student mathematics achievement, as measured by the LEAP assessment and relative to students who do not receive any form of Zearn tutoring?
2. What is the impact of using Zearn Supplemental with dedicated implementation support on student mathematics achievement, as measured by the LEAP assessment and relative to students who receive standard Zearn tutoring (not the special models used in the three target districts)?
3. How do the impacts of Zearn Supplemental with dedicated implementation support vary by student characteristics (e.g., grade, race, ethnicity)?
4. What is the effect of Zearn Supplemental with dedicated implementation support on student outcomes for students who received the full treatment (defined as 90 lessons)?
5. To what extent is greater fidelity of implementation (average weekly Zearn usage) associated with better student outcomes?

METHOD

Research Design

This study examined the impacts of Zearn Supplemental with dedicated implementation support (“Zearn Supplemental”) by using a quasi-experimental design (QED) with matching across treatment and comparison conditions. Specifically, we examined the mathematics achievement of 4th-8th grade students in the State of Louisiana during the 2023-24 school year. The study was designed to meet the What Works Clearinghouse (WWC) evidence standards “with reservations” and qualify as *Tier 2* evidence according to the Every Student Succeeds Act (ESSA)¹, where schools in the three school districts implemented Zearn Supplemental while other, similar schools in demographically comparable districts in the state employed other (“business-as-usual”) approaches.

Three districts (Lincoln, Natchitoches, and St. John the Baptist Parish districts) were used as the treatment group as these districts had agreed to implement Zearn with fidelity and use the supplemental Zearn implementation support services. However, because all Louisiana districts had access to or licenses for using the Zearn program from the State of Louisiana, it was important to identify and define an appropriate comparison group for analysis. Ultimately, a definition of non-user districts was developed based on an examination of school-level Zearn usage from the 2022-23 school year. In reviewing these prior Zearn usage trends, it was determined that a school that on average completed fewer than one lesson per week was a meaningful cutoff point. This allowed for the assumption that in those schools, while there may have

¹ A study may meet WWC Tier 2 standards if it is a strong quasi-experimental study (one without random selection and/or assignment to treatment/comparison conditions), with similar comparison groups at baseline, includes at least 350 students and at least two educational sites, and has statistically significant positive effects.



been a grade or teacher that was using Zearn on a more consistent basis, there was no systemic adoption or implementation of Zearn at the district level. This ensured there was a distinctive difference in usage compared to the three treatment districts.

In the first grouping (termed Comparison Group 1), the three treatment districts were matched with districts that were defined as non-Zearn users. For this analysis, given the wide access to Zearn across the state, we defined non-use as districts in which schools on average completed less than one lesson per week during the prior academic year. In the second grouping (Comparison Group 2), the three treatment districts were matched with districts where students received Zearn tutoring without implementation support. For this analysis, Zearn usage without implementation supports was defined as districts that did not receive the Zearn Supplemental additional supports, such as school-level coaching, and in which schools on average completed more than one lesson per week in prior usage. In the third matched grouping (Comparison Group 3), the three Zearn Supplemental districts were matched with a subset of districts identified by Zearn that intended to fully implement Zearn district wide. The secondary and tertiary analyses were intended to reveal the value-added benefit for achievement, if any, of the implementation support programming.

To create Comparison Group 1, we dropped from the comparison district matching pool any districts with fewer than eight or more than 50 schools. In addition, Jefferson Parish was removed from the research sample at the request of Zearn. Any schools with only K-3 enrollment, as well as schools missing data for variables of interest (DPS score, K-8 Progress Index score, Letter Grade) were also removed. Two schools from Natchitoches Parish (N.S.U. Elementary Lab School and N.S.U. Middle Lab School) and one school from Lincoln Parish were dropped as they opted out of participation in the treatment program prior to the start of the study.

To create Comparison Group 2, a similar procedure was followed. Jefferson Parish and the schools in Lincoln and Natchitoches Parishes were dropped, as were K-3 only schools and schools with missing data for relevant variables of interest. In addition, six parishes (Ascension, Lafayette, Acadia, St. Charles, St. Landry, and Terrebonne) were dropped because these school districts never intended to implement Zearn. Once more, districts that had fewer than eight schools or more than 50 schools were dropped. Districts without prior Zearn usage data were also dropped because we would be unable to match them appropriately in these research questions, which focused on comparing regular users with coaching to regular users without coaching. Therefore, for the purposes of matching, treatment was defined as districts that received coaching.

Based on these assumptions, two groupings for Comparison Groups 1 and 2 were created, and one-to-one multi-level matching (MLM) was used to create district and school matches. In the first round of MLM, schools were nested within district codes, so that schools within matched districts were also matched across districts. Fine balance constraints were used at the district level, applying categorical stratification to ensure balance across district-level characteristics. We required 100% of treated schools to be kept in the matched sample.



Matching for Comparison Group 1 (comparing Zearn users with non-users) was based on school prior achievement (proportion of students meeting Mastery level or higher on the Spring 2023 LEAP assessment), school demographic enrollment rate (enrollment rate for Black or African American students), district economic characteristics (proportion of economically disadvantaged students), and district average school progress (K-8 Progress Index scores). Matching for Comparison Group 2 (comparing typical Zearn users who also received coaching with typical Zearn users without coaching) was based on prior school Zearn usage (average weekly lessons completed), school prior achievement, proportion of active Zearn users, as well as district economic characteristics and district economic enrollment rate (enrollment rate for Black or African American students). In both matching procedures, to reduce the influence of outliers, improve covariate balance, and facilitate matching, two-level cut scores were calculated for district-level variables. For Comparison Groups 1 and 2, matching was conducted three times to create 1:3 matching. This allowed us to overmatch each district in case districts opted not to complete data sharing agreements with the LDOE.

To create Comparison Group 3, Zearn identified districts that had intended to implement Zearn fully across their entire district. From those districts, districts were removed that met the above criteria related to size and data availability. This set of districts were considered the pool for Comparison Group 3 and no further matching was required.

To conduct the impact analysis at the student level, a second round of multi-level matching was conducted across districts, using school and student-level data (student-level variables: prior LEAP achievement, grade, and race/ethnicity, and school-level variables: prior LEAP achievement, school enrollment, and district pair IDs from the first round of matching for Comparison Groups 1 and 2). To reduce the influence of outliers, improve covariate balance, and facilitate matching, two-level cut scores were calculated for school-level variables. Prior to matching, students with no grade in the 2023-24 school year were removed from the data. No grade indicated that the student had been enrolled in the prior year but was not enrolled in the year of implementation. While we did not remove any students from the pool based on their actual Zearn usage, we did require that they were enrolled in the school for the year of implementation. In each of the district groupings, students and schools in the treatment districts were matched with students and schools in the appropriate comparison group districts.

Participants

Details about study participants are presented below. To answer the research questions stated above, three comparison groups were created through statistical matching processes. The State of Louisiana consists of 69 school districts, serving approximately 676,751 students across 1,367 schools as of September 2024. Three districts, Lincoln, Natchitoches, and St. John the Baptist Parishes (a term used instead of county and is without religious connotations), previously identified as sites for the Zearn Supplemental intervention, served as the treatment sites for this study in each comparison group.

Lincoln Parish Schools encompasses 15 schools in North Louisiana. As of the 2023-24 school year, 5,809 students were enrolled. The district is 40% Black, 8% Latino/a/e,



49% White, and 3% other races. At the district level, 33.8% of families receive Food Stamp/SNAP benefits. In total, 2.8% of students are English Language Learners, and 9.5% of students are categorized as having disabilities.

Natchitoches Parish Schools encompasses 13 schools in Central Louisiana. As of the 2023-24 school year, 4,947 students were enrolled. The district is 56% Black, 4% Latino/a/e, 35% White, and 5% other races. At the district level, 33.2% of families receive Food Stamp/SNAP benefits. In total, 1.2% of students are English Language Learners, and 6.4% of students are categorized as having disabilities.

St. John the Baptist Parish Schools encompasses 11 schools in the Greater New Orleans region of Louisiana. As of the 2023-24 school year, 5,202 students were enrolled. The district is 72% Black, 16% Latino/a/e, 9% White, and 3% other races. At the district level, 40% of families receive Food Stamp/SNAP benefits. In total, 7.5% of students are English Language Learners, and 10.3% of students are categorized as having disabilities.

The State of Louisiana Department of Education (LDOE) requested each district identified as a potential match for one of the three comparison groups described above sign a data sharing agreement to share their student-level achievement and demographic data with the Johns Hopkins CRRE study team. After 14 districts signed data sharing agreements, students and schools for each comparison group were matched through multi-level matching based on prior student- and school-level achievement, race/ethnicity, grade, school enrollment, and district matched pairs from the initial district matching. This resulted in an analytic sample of 7,142 in the first comparison group, 7,006 in the second comparison group, and 5,756 in the third comparison group. Below, we report demographic data for students in each analytic group. A full consideration of baseline equivalence will be presented in the results section for each comparison group.

Comparison Group 1: Zearn Supplemental vs. non-Zearn Districts



26 treatment schools



2,538 tutored students
in Grades 4-8

**Table 1***Student characteristics for Comparison Group 1 analytic sample, by treatment group*

		Treatment N (%)	Matched Comparison N (%)	Overall N (%)
Grade	4	11 (0.2)	6 (0.2)	11 (0.2)
	5	1045 (20.7)	482 (19.1)	1045 (20.7)
	6	1358 (26.8)	679 (26.9)	1358 (26.8)
	7	1514 (29.9)	805 (31.9)	1514 (29.9)
	8	1132 (22.4)	550 (21.8)	1132 (22.4)
Race/Ethnicity	Black	1443 (56.9)	1403 (55.6)	2846 (56.2)
	Latino/a/e	216 (8.5)	177 (7.0)	393 (7.8)
	Other Race/Ethnicity	78 (3.1)	170 (6.7)	248 (4.9)
	White	801 (31.6)	772 (30.6)	1573 (31.1)
English Language Learner Status	Yes	62 (2.4)	71 (2.8)	133 (2.6)
	No	2476 (97.6)	2451 (97.2)	4927 (97.4)
Special Education Status	Yes	287 (11.3)	318 (12.6)	605 (12.0)
	No	2251 (88.7)	2204 (87.4)	4455 (88.0)
Economically Disadvantaged Status	Yes	2040 (80.4)	2066 (81.9)	4106 (81.1)
	No	498 (19.6)	456 (18.1)	954 (18.9)
<i>n</i>		2538	2522	5060

Comparison Group 1 included Ascension, Livingston, Ouachita, Terrebonne, Acadia, and St. Landry Parishes. Across these nine districts, student sample composition was similar across conditions, with no standardized mean differences between groups greater than 0.25 standard deviations. A majority of students (>50%) were Black or African American, followed by White (31%) and Latino/a/e (7.8%). Approximately 80% were classified as economically disadvantaged, while about 11% were identified as receiving special education services, and 2% were identified as English Learners. Students in Zearn Supplemental districts were slightly more likely to receive special education services, and slightly less likely to be English Learners and economically disadvantaged, though not meaningfully so.

Comparison Group 2: Zearn Supplemental vs. Zearn User Districts without Implementation Support





25 treatment schools 2,576 tutored students
in Grades 4-8

The second sample compared students from the three Zearn Supplemental districts with Zearn districts without implementation supports such as school-level coaching: Bossier, Vermilion, Lafourche, and St. Mary Parish, as well as City of Monroe School District. The student sample was similar across conditions, with no standardized mean differences between groups greater than 0.25 standard deviations. As in the first comparison group, the majority of these students were Black or African American (>50%) or White (30%). Students were majority economically disadvantaged (>80%), while not receiving special education services or English Learner supports.

Table 2

Student characteristics for Comparison Group 2 analytic sample, by treatment group

		Treatment N (%)	Matched Comparison N (%)	Overall N (%)
Grade	4	14 (0.3)	9 (0.4)	14 (0.3)
	5	1379 (27.5)	709 (29.1)	1379 (27.5)
	6	1274 (25.4)	612 (25.1)	1274 (25.4)
	7	1107 (22.1)	443 (18.2)	1107 (22.1)
	8	1238 (24.7)	663 (27.2)	1238 (24.7)
Race/Ethnicity	Black	1526 (59.2)	1388 (57.0)	2914 (58.1)
	Latino/a/e	216 (8.4)	148 (6.1)	364 (7.3)
	Other Race/Ethnicity	69 (2.7)	157 (6.4)	226 (4.5)
	White	765 (29.7)	743 (30.5)	1508 (30.1)
English Language Learner Status	Yes	57 (2.2)	52 (2.1)	109 (2.2)
	No	2519 (97.8)	2384 (97.9)	4903 (97.8)
Special Education Status	Yes	289 (11.2)	331 (13.6)	620 (12.4)
	No	2287 (88.8)	2105 (86.4)	4392 (87.6)
Economically Disadvantaged Status	Yes	2096 (81.4)	1965 (80.7)	4061 (81.0)
	No	480 (18.6)	471 (19.3)	951 (19.0)
<i>n</i>		2576	2436	5012

Comparison Group 3: Zearn Supplemental vs. Zearn-Identified Similar Zearn Math Districts





25 treatment schools 2,256 tutored students
in Grades 4-8

The third comparison group sample compared the three Zearn Supplemental districts with three districts identified by Zearn as a district with similar implementation to the treatment districts: Bossier and Lafourche Parishes, and City of Monroe School District. Across the six districts, the student sample was similar across conditions, with no standardized mean differences between groups greater than 0.25 standard deviations. As in the first two comparison groups, most of these students were Black or African American (>50%) or White (>30%). Students were majority economically disadvantaged (>80%), while not receiving special education services or English Learner supports.

Table 3

Student characteristics for Comparison Group 3 analytic sample, by treatment group

		Treatment N (%)	Matched Comparison N (%)	Overall N (%)
Grade	4	7 (0.3)	14 (0.6)	21 (0.5)
	5	648 (28.7)	861 (37.3)	1509 (33.1)
	6	581 (25.8)	848 (36.8)	1429 (31.3)
	7	528 (23.4)	340 (14.7)	868 (19.0)
	8	492 (21.8)	243 (10.5)	735 (16.1)
Race/Ethnicity	Black	1236 (54.8)	1237 (53.6)	2473 (54.2)
	Latino/a/e	165 (7.3)	157 (6.8)	322 (7.1)
	Other Race/Ethnicity	75 (3.3)	120 (5.2)	195 (4.3)
	White	780 (34.6)	792 (34.3)	1572 (34.5)
English Language Learner Status	Yes	38 (1.7)	59 (2.6)	97 (2.1)
	No	2218 (98.3)	2247 (97.4)	4465 (97.9)
Special Education Status	Yes	254 (11.3)	314 (13.6)	568 (12.5)
	No	2002 (88.7)	1992 (86.4)	3994 (87.5)
Economically Disadvantaged Status	Yes	1784 (79.1)	1738 (75.4)	3522 (77.2)
	No	472 (20.9)	568 (24.6)	1040 (22.8)
<i>n</i>		2256	2306	4562

Measures

In order to address the research questions, the study team gathered and analyzed data including student achievement data (LEAP mathematics assessment), school- and student-level demographic data, and Zearn usage data (see Table 4).

The outcome measure for the study included student performance on the LEAP assessment of mathematics achievement, which is given to students in Grades 3-8. LEAP is aligned to the Louisiana Student Standards and is given every spring.



We obtained from Zearn available data regarding program usage at each school and by each student. Zearn shared school-level data focused on prior usage for the purposes of conducting district and school matching. Student-level usage data was shared after the end of the 2023-24 school year. These datasets were shared with the Louisiana Department of Education, who then shared all student-level data, including usage data, via a secure FTP. Relevant usage metrics for the matching included average number of active Zearn users and average number of lessons completed. Relevant student-level usage metrics included total number of lessons completed; number of unique below, on, and above grade-level lessons completed; minutes of Zearn usage; and first and last sign-in date.

Table 4
Research questions with data sources and measures

Research questions	Student achievement data	School-level demographic data	Student-level demographic data	Zearn usage data
1. What is the impact of using Zearn Supplemental with dedicated implementation support on student mathematics achievement, as measured by the LEAP assessment and relative to students who do not receive any form of Zearn tutoring?	✓	✓	✓	
2. What is the impact of using Zearn Supplemental with dedicated implementation support on student mathematics achievement, as measured by the LEAP assessment and relative to students who receive standard Zearn tutoring (not the special models used in the three target districts)?	✓	✓	✓	
3. How do the impacts of Zearn Supplemental with dedicated implementation support vary by student characteristics (e.g., grade, race, ethnicity)?	✓	✓	✓	
4. What is the effect of Zearn	✓	✓	✓	✓



Supplemental with dedicated implementation support on student outcomes for students who received the full treatment (defined as 90 lessons)?				
5. To what extent is greater fidelity of implementation associated with better student outcomes?	✓	✓	✓	✓

Data sources and measures for the current study are described below.

Louisiana Educational Assessment Program (LEAP) Mathematics Assessment.

The LEAP mathematics assessment is a statewide standardized mathematics assessment for students in Grades 3-8, aligned to the Louisiana Student Standards for Mathematics (LDOE, 2025). The test is conducted in both online and paper-based formats in the spring of each school year. Despite the timing of the testing at the end of the year, the Spring 2023 assessment will be referred to as a pretest and serve as the baseline measure, while the Spring 2024 assessment will be referred to as the posttest and serve as the outcome measure. All analyses were conducted using scale scores. Scoring for the assessment is not vertical, with scores covering the same range in each grade, meaning that a score of 300 in Grade 4 is not equivalent to a score of 300 in Grade 6. To account for this, student grade was used as a covariate in the analytic models.

Zearn Math Usage data. Zearn provided digital usage data to the Louisiana Department of Education. Student-level usage data was then shared with CRRE from the districts with signed DSAs. Prior usage data from 2022-23 school year were available at the school level for matching purposes. Usage data at the individual level were available for analytic purposes. Data about lessons included: total lessons completed; number of unique below, on, and above-grade-level lessons completed; minutes of usage; and weeks of usage.

Demographics (District, School, and Student). District-level and school-level aggregate data were available publicly on the LDOE website. District-level variables included in the matching were prior achievement, Black or African American enrollment, and school enrollment. School-level variables included in the analytic models were school enrollment, proportion of Black or African American students in a school, the district in which a school is located, and the percentage of students in Grades 4-8 reaching Mastery or higher on LEAP. The State of Louisiana provided student-level demographic data. Student variables included in the analytic models were race/ethnicity (Latino/a/e, Black, White, or other race), special education status (SPED), English Learner status (ELL), and poor status (FRL).

Analytical Approach

All analyses were completed using the *R* quantitative analysis software (version



2024.12.1; R Core Team, 2024). We used hierarchical linear modeling (HLM) through the *lmerTest* package (Kuznetsova et al., 2017) with students nested within schools to examine differences in spring 2024 LEAP mathematics achievement between Zearn Supplemental with dedicated implementation support treatment and comparison students, adjusting for prior achievement and other demographic covariates. Schools were chosen as the clustering variable for Level 2, as Zearn Supplemental usage was intended to be determined at the school (rather than classroom) level. To facilitate interpretation of results for non-academic readers, the LEAP scale scores were left in their original scale, and grade was controlled for in all models. The models account for matching through conditioning on the variables used in the matching process and therefore do not need to include the propensity scores in the model (Stuart, 2010). The final HLM model used to estimate the impact of treatment on mathematics achievement was:

$$Y_{ij} = \gamma_{00} + \gamma_{10}(StudentPretest)_{ij} + \gamma_{K0}W_{ij} + \gamma_{01}(Treatment)_j + \gamma_{02}(District)_j + \gamma_{0M}V_j + u_{0j} + r_{ij}$$

where Y_{ij} is the LEAP mathematics score in spring 2024 for student i in school j ; γ_{00} is the covariate-adjusted grand mean test score for the comparison group; γ_{10} is the regression coefficient for the student pretest (LEAP mathematics score in spring 2023); γ_{K0} is a vector of regression coefficients for student covariates; W_j is a vector of student demographic and grade level covariates (grade level, race/ethnicity, ELL status, socioeconomic status, and special education status); γ_{01} is the average treatment effect; $(Treatment)_j$ is the binary treatment indicator for school j ; γ_{02} is the regression coefficient for the district indicator; $(District)_j$ is the district indicator; γ_{0M} is a vector of regression coefficients for school variables; V_j is a vector of school covariates (school overall enrollment, school-level Black or African American enrollment proportion, and the proportion of students in the school reaching Mastery or higher in prior academic year); and u_{0j} is the random school effect for school j ; r_{ij} is the student-level residual error term. All continuous covariates were grand mean centered to facilitate interpretation of the intercept.

As part of the design of this quasi-experiment, we matched treatment and control students using propensity score matching. As described above, districts and schools were previously matched using *R* package *matchMulti* (Keele et al., 2024) based on school and district-level variables, to form two of the comparison groups, while one set of districts was identified by Zearn (described above). Subsequently, students from the districts with signed DSAs with the LDOE were matched across districts and schools, using student-level and school-level variables (described above). For a full explanation of the flow of participants, please see the consort diagram in Appendix A and the attrition analyses in Appendix B.

Procedure

Zearn collaborated with the LDOE and parish school districts to identify districts interested in implementing Zearn Supplemental with dedicated implementation support



with fidelity, meaning they would work to ensure students completed three lessons per week and the schools in the district participated in school-level matching. Other districts in Louisiana had the opportunity to implement Zearn without implementation support.

School- and district-level data were available and collected from public websites run by the LDOE. Student-level achievement and demographic data were made available by the LDOE to the researchers for students from the districts with signed DSAs with the LDOE. Zearn provided student usage data for students from DSA-signed districts to the LDOE, which then shared the data with CRRE.

RESULTS

This section of the report provides the findings from each of the research questions, organized by comparison group. This section begins with the findings related to Comparison Group 1, comparing Zearn Supplemental students with non-users (students from districts where in prior years students completed less than one lesson per week). The next section describes the findings pertaining to Comparison Group 2, comparing Zearn supplemental students with Zearn users in districts without implementation support. Finally, we present results pertaining to Comparison Group 3, comparing Zearn Supplemental students with those from Zearn-identified intended users. In each section, we describe the impact analysis [RQ1 or RQ2], how these effects vary by student characteristics [RQ3], the effect of Zearn on student outcomes for those who received the full treatment (defined as 90 lessons) [RQ4], and an examination of the extent to which greater fidelity of implementation was associated with better student outcomes (that is, is 90 lessons indeed the true tipping point for meaningful impact) [RQ5].

Comparison Group 1: Impact of Zearn Supplemental Compared to non-Zearn Users

Key Findings

- > Students using Zearn Supplemental scored 5.97 points higher than non-Zearn users on the LEAP mathematics assessment ($p < .05$), with an effect size of +0.20 SD.
- > No differential effects by student characteristics were identified. Students using Zearn Supplemental had similar outcomes across race/ethnicity groups, English learner status, economic disadvantage status, and special education participation.
- > Students completing ≥ 90 lessons (regardless of condition) outscored those with lower usage by 4.7 points ($p < .05$; effect size +0.15 SD).



Students with high levels of usage (lessons completed and average lessons completed per week) outscored those with negligible or lower usage levels.

This section describes findings related to Comparison Group 1, comparing the effects of Zearn Supplemental against non-Zearn users. Because Zearn could be implemented in other districts without school-level coaching, we defined non-users as those schools where the average prior usage was less than one lesson per week.

Tables 5 and 6 describe the school- and student-level characteristics, respectively, of Comparison Group 1 prior to and after matching. Prior to matching, 52 comparison schools with 12,324 students and 26 treatment schools with 4,740 students were identified with non-missing spring 2023 LEAP Math scores and demographic and usage data. At that stage, school- and student- level differences existed across multiple variables, including school attendance, Black or African American enrollment rate, prior mathematics achievement, grade, student race/ethnicity, English Learner status, and special education status. Thus, propensity score matching (PSM) was used to create a comparison group of students that was as similar as possible on observable factors to treatment students.

Propensity scores were computed using the *matchMulti* package (Keele et al., 2024) command in *R* (R Core Team, 2024) (v 4.4.3); multilevel matching for data with cluster level treatments and individual-level outcomes using a network optimization algorithm was used to create the sample. The result of this matching procedure was that treatment students within treatment schools were matched across districts with comparison students in comparison schools who were as similar as possible in terms of prior LEAP achievement and demographic variables, allowing for a more rigorous contrast of treatment and comparison students. The matched samples demonstrated baseline equivalence overall and across all grade levels.

The final Comparison Group 1 analytic sample at baseline, also described in the rightmost columns in Tables 5 and 6, consisted of 2,538 treatment students and 2,522 comparison students across 24 treatment and 23 comparison schools. The final column shows the standardized mean differences between treatment and comparison group schools and students. Most importantly, schools and students were similar in their prior academic achievement and in most demographic variables, as indicated by the standardized mean differences being lower than 0.25 SDs. Comparison schools had somewhat higher school enrollment (417 vs. 377 students, SMD = 0.23) and lower prior rates of active Zearn users (0.43 vs. 0.52, SMD = 0.31), though, as described above, this was by design.



Table 5
Comparison Group 1 school level baseline equivalence, prior to and post-matching, final analytic sample

Category	Level	Pre-Matching Comparison	Pre- Matching Treatment	Post- Matching Comparison	Post- Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
<i>n</i>		52	26	26	26	23	24	
School Attendance Rate Mean (SD)		93.57 (1.62)	91.91 (4.81)	92.77 (1.26)	91.91 (4.81)	92.89 (1.28)	92.76 (2.19)	0.074 (0.80)
% of Students 2022- 2023 LEAP Mastery+ Grade 3-8 Mean (SD)		31.50 (18.55)	24.46 (20.33)	25.81 (15.47)	24.46 (20.33)	26.09 (15.65)	26.12 (20.29)	0.002 (0.99)
Total Enrollment Mean (SD)		422.56 (176.14)	360.27 (166.70)	416.46 (179.36)	360.27 (166.70)	417.09 (177.06)	377.38 (161.68)	0.234 (0.43)
LEP Enrollment Rate Mean (SD)		0.04 (0.04)	0.03 (0.04)	0.04 (0.04)	0.03 (0.04)	0.04 (0.05)	0.03 (0.04)	0.264 (0.37)
Economically Disadvantaged Rate Mean (SD)		0.76 (0.16)	0.81 (0.20)	0.79 (0.15)	0.81 (0.20)	0.79 (0.15)	0.80 (0.20)	0.047 (0.87)
Black/African American Enrollment Rate Mean (SD)		0.41 (0.30)	0.57 (0.28)	0.45 (0.31)	0.57 (0.28)	0.46 (0.31)	0.54 (0.27)	0.271 (0.36)
White Enrollment Rate Mean (SD)		0.44 (0.28)	0.32 (0.30)	0.39 (0.28)	0.32 (0.30)	0.37 (0.27)	0.35 (0.30)	0.093 (0.75)
Latino/a/e Enrollment Rate Mean (SD)		0.09 (0.08)	0.08 (0.07)	0.09 (0.08)	0.08 (0.07)	0.09 (0.08)	0.08 (0.07)	0.101 (0.73)
Grades 4-8 Enrollment Mean (SD)		247.48 (164.05)	203.08 (136.22)	292.27 (166.99)	203.08 (136.22)	293.35 (143.71)	218.04 (130.87)	0.548 (0.07)
Proportion of Active Zearn Users in School (Prior year) Mean (SD)		0.30 (0.30)	0.49 (0.25)	0.39 (0.35)	0.49 (0.25)	0.43 (0.36)	0.52 (0.23)	0.308 (0.30)

Table 6



Comparison Group 1 student level baseline equivalence, prior to and post-matching, final analytic sample

Category	Level	Pre-Matching Comparison	Pre-Matching Treatment	Post-Matching Comparison	Post-Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
n		12324	4740	3571	3571	2522	2538	
2022-2023 LEAP Math Level N (%)	Advanced	395 (3.2)	164 (3.5)	74 (2.1)	139 (3.9)	46 (1.8)	98 (3.9)	
	Mastery	3505 (28.4)	1087 (22.9)	841 (23.6)	796 (22.3)	620 (24.6)	600 (23.6)	
	Basic	3544 (28.8)	1244 (26.2)	950 (26.6)	961 (26.9)	724 (28.7)	739 (29.1)	
	Approaching Basic	3268 (26.5)	1431 (30.2)	1089 (30.5)	1064 (29.8)	757 (30.0)	751 (29.6)	
	Unsatisfactory	1612 (13.1)	814 (17.2)	617 (17.3)	611 (17.1)	375 (14.9)	350 (13.8)	
2022-2023 LEAP Math Score Mean (SD)		734.39 (30.63)	729.84 (32.01)	728.98 (30.28)	730.10 (32.21)	730.35 (29.02)	732.12 (30.88)	0.059 (0.04)
Race/Ethnicity N (%)	Black or African American	4691 (38.1)	2864 (60.4)	2038 (57.1)	2048 (57.4)	1403 (55.6)	1443 (56.9)	
	Latino/a/e	1131 (9.2)	412 (8.7)	222 (6.2)	303 (8.5)	177 (7.0)	216 (8.5)	
	Other	1006 (8.2)	145 (3.1)	211 (5.9)	107 (3.0)	170 (6.7)	78 (3.1)	
	White	5496 (44.6)	1319 (27.8)	1100 (30.8)	1113 (31.2)	772 (30.6)	801 (31.6)	
Economically Disadvantaged Status N (%)	Y	9116 (74.0)	3864 (81.5)	2904 (81.3)	2888 (80.9)	2066 (81.9)	2040 (80.4)	
	N	3208 (26.0)	876 (18.5)	667 (18.7)	683 (19.1)	456 (18.1)	498 (19.6)	
English Learner (EL) Status Mean (SD)	Y	425 (3.4)	130 (2.7)	88 (2.5)	87 (2.4)	71 (2.8)	62 (2.4)	
	N	11899 (96.6)	4610 (97.3)	3483 (97.5)	3484 (97.6)	2451 (97.2)	2476 (97.6)	
Special Education Status Mean (SD)	Y	2218 (18)	948 (20)	607 (17)	714 (20)	429 (17)	508 (20)	
	N	10106 (82)	3792 (80)	2964 (83)	2857 (80)	2093 (83)	2030 (80)	



Research Question 1: Impact Analysis

This first analysis depicts the adjusted and unadjusted descriptive trends in LEAP scores from EOY to EOY (as LEAP is administered each spring) for treatment and matched comparison students in the 2023-24 school year. Table 7 depicts the unadjusted and adjusted means for the treatment and comparison groups at baseline and posttest, as well as the results from the impact analysis.

Table 7

Impact analysis results and mean scores (pretest and posttest), by treatment group

Time	Unadjusted Comparison Mean (SD)	Unadjusted Treatment Mean (SD)	Adjusted Comparison Mean	Adjusted Treatment Mean	Impact Estimate (SE)	p Value	Effect Size
Pretest	730.35 (29.02)	732.12 (30.88)					
Posttest	728.68 (29.83)	733.06 (32.58)	727.82	733.79	5.97 (2.78)	.040	+0.20

This multiple regression analysis controlled for 2022-23 EOY LEAP Math scores and included student and school-level demographic variables. All continuous covariates were grand mean centered to enable interpretation of the intercept. Trends are shown for all students in the analytic sample. The full model can be found in Appendix C.

The Zearn Supplemental program was shown to have a statistically significant ($p = .040$) positive impact on LEAP scores. The regression estimate can be interpreted as the average difference in spring 2024 LEAP mathematics scores between treatment and matched comparison students. Thus, on average, treatment students outscored matched comparison students (non-Zearn users) by 5.97 points on the spring LEAP assessment, controlling for prior achievement and demographic variables. The effect size of .20 SDs is indicative of a large practical impact of Zearn on LEAP scores (Kraft, 2020).

Research Question 3: Subgroup Analysis

The next section presents the effects of the Zearn Supplemental intervention for different student subgroups. Table 8 displays the group means for the two-level model, with students within schools, while the full model with interactions is shown in Appendix C. Differential effects were measured based on grade, race/ethnicity, economic disadvantaged status, special education status, and English Language Learner status. No differential effects were identified by student characteristic. Differential effects were identified by grade with somewhat larger impacts in 8th grade. However given the nature of the scale, we choose to report the adjusted means by condition in the scale of the test for clearer interpretation by practitioners familiar with the LEAP assessment.

**Table 8**

Subgroup analysis results, Zearn impacts on LEAP math scores: Group means

Category	Sub-Category	Comparison			Treatment		
		Estimate	SE	p value	Estimate	SE	p value
Race/Ethnicity	Black or African-American	726.43	1.61	<.001	732.55	1.55	<.001
	Latino/a/e	730.65	2.10	<.001	736.96	1.92	<.001
	Other	730.09	1.96	<.001	734.21	2.32	<.001
	White	729.27	1.70	<.001	735.17	1.62	<.001
Economically Disadvantaged Status	Y	727.23	1.58	<.001	733.44	1.51	<.001
	N	730.36	1.75	<.001	735.30	1.71	<.001
Special Education Status	N	728.29	1.58	<.001	734.27	1.51	<.001
	Y	724.36	1.78	<.001	730.23	1.75	<.001
English Language Learner Status	ELL	728.25	2.74	<.001	732.69	2.74	<.001
	not ELL	727.81	1.57	<.001	733.82	1.50	<.001
Grade	5	727.88	1.75	<.001	732.83	1.66	<.001
	6	728.28	1.66	<.001	732.55	1.61	<.001
	7	729.58	1.66	<.001	733.62	1.62	<.001
	8	724.58	1.74	<.001	736.20	1.65	<.001

Note. While 4th grade was included in the model, the sample size for 4th grade was so small that we chose not to include those results.

Research Question 4: Usage Analysis

In the next section, we examine the effect of Zearn Supplemental on student outcomes for those who received the full treatment, defined as 90 unique on-grade-level lessons over the course of the treatment period. As shown in Table 9, there were 5,060 individual students with Zearn usage data across treatment and comparison conditions in Comparison Group 1. As would be expected, given the opt-in process to participate in Zearn Supplemental and district- and school-level matching, students in treatment districts completed a greater number of unique lessons and were more likely to have completed an adequate number of levels.

**Table 9***Descriptive statistics on usage by condition*

Category	Level	Overall	Comparison	Treatment	<i>p</i>	SMD
<i>n</i>		5060	2522	2538		
Total Unique On-Grade-Level Lessons Mean (SD)		37.91 (48.75)	4.84 (15.91)	70.78 (48.16)	<0.001	1.839
Zearn Usage Level <i>N</i> (%)	Inadequate	4105 (81.1)	2510 (99.5)	1595 (62.8)		
	Adequate	955 (18.9)	12 (0.5)	943 (37.2)		

Note. Adequate usage is defined as completing 90 unique lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique lessons.

In the analysis of the effect of Zearn Supplemental on student outcomes for those who received the full treatment, hierarchical linear models (HLM) were used, with students nested within schools as in the previous analyses. However, the treatment variable was replaced with student usage as the independent variable. As in the previous analytic models, student spring 2024 LEAP mathematics scores were the dependent variable. The mean scores at pretest and posttest are described in Table 10. These groups differed significantly at baseline, with students' pretest scores 0.49 standard deviations different. To put this in perspective, What Works Clearinghouse states that an effect size for a baseline characteristic between 0.05 and 0.25 necessitates a statistical adjustment (WWC, 2024). Anything higher than that in absolute value indicates that the groups are not equivalent at baseline. These findings should be considered exploratory and suggestive.

Table 10*Impact of usage on achievement, with mean scores (pretest and posttest), by usage level (2-level)*

Timepoint	Unadjusted Inadequate (<90) Mean (SD)	Unadjusted Adequate (>=90) Mean (SD)	Adjusted Inadequate (<90) Mean	Adjusted Adequate (>=90) Mean	Impact Estimate (SE)	<i>p</i> Value	Effect Size
Pretest	728.46 (29.84)	743.17 (27.54)					+0.49
Posttest	727.92 (31.33)	743.58 (27.88)	729.98	734.72	4.74 (0.86)	<.001	+0.15

Note. Adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique on-grade-level lessons.

The full HLM model is shared in Appendix C. Findings show that students receiving the full treatment outscored students who completed fewer than 90 lessons by 4.74 points on the spring LEAP assessment, controlling for prior achievement and demographic



variables. The effect size of .15 SDs is indicative of a medium practical impact of Zearn Supplemental on LEAP scores (Kraft, 2020).

Research Question 5: Exploratory Usage Analysis

Finally, we conducted an exploratory analysis with each dataset in which we examined whether the designed full intervention of 90 lessons was indeed the tipping point for impact. To do so, we examined different usage variables as provided by Zearn through the Louisiana Department of Education. Descriptive usage statistics for Comparison Group 1 are described in Table 11.

Zearn provided variables counting the total number of lessons completed; the total number of unique lessons completed below grade level, on grade level, and above grade level; and weeks of Zearn usage. Using these variables, we also calculated the total number of unique lessons completed (adding together their below, on grade level, and above grade level unique lessons) and the average number of lessons per week (dividing total lessons by weeks of Zearn usage). In addition, we created a four-level categorical variable using cutoffs from Zearn, and a two-level categorical variable based on the 90 lessons threshold. The two-level cutoff used the 90 lessons completed as the cutoff point, as in the previous analysis. In the four-level cutoff model, the levels were defined as: Negligible (fewer than 30 lessons completed), Low Usage (between 30 and 59 lessons completed), Medium Usage (60-89 lessons completed), and High Usage (90 or more lessons completed).

As shown in Table 11, students in the treatment group districts completed more total lessons, more unique lessons, and completed more lessons per week. Because of these differences in prior usage, the findings from this analysis cannot be considered causal.

Table 11
Descriptive statistics on usage by condition

Category	Level	Overall	Comparison	Treatment	<i>p</i>	SMD
<i>n</i>		5060	2522	2538		
Total Unique Below-Grade Level Lesson Completions Mean (SD)		1.04 (5.50)	0.33 (3.12)	1.73 (7.05)	<0.001	0.256
Total Unique Above-Grade Level Lesson Completions Mean (SD)		1.01 (7.00)	0.07 (1.07)	1.94 (9.74)	<0.001	0.270
Total Unique On-Grade Level Lesson Completions Mean (SD)		35.87 (45.26)	4.44 (14.79)	67.11 (43.70)	<0.001	1.921
Total Lesson		42.54	5.05 (16.86)	79.78	<0.001	1.776



Completions (including repeated lessons) Mean (SD)		(56.32)		(57.07)		
Total Unique Lesson Completions Mean (SD)		37.91 (48.75)	4.84 (15.91)	70.78 (48.16)	<0.001	1.839
Average Lessons Completed Per Week Mean (SD)		1.36 (1.49)	0.34 (0.73)	2.20 (1.43)	<0.001	1.642
Zearn Usage Level (2-levels) N (%)	Inadequate	4105 (81.1)	2510 (99.5)	1595 (62.8)		
	Adequate	955 (18.9)	12 (0.5)	943 (37.2)		
	Negligible	3027 (59.8)	2395 (95.0)	632 (24.9)		
Zearn Usage Level (4-levels) N (%)	Low Usage	514 (10.2)	62 (2.5)	452 (17.8)		
	Medium Usage	564 (11.1)	53 (2.1)	511 (20.1)		
	High Usage	955 (18.9)	12 (0.5)	943 (37.2)		

Note. In the 2-Level model, adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique lessons. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

To conduct this analysis, we explored usage in multiple ways. First, we used the four-level categorical approach using cutoffs from Zearn in our analytic model. In this first model, the four-level usage variable was used as the independent variable, and student LEAP Math achievement score was once more the dependent variable. As can be seen in Table 12, which describes the unadjusted mean pretest and posttest scores for students in each of the four usage categories, students with high usage performed higher on the Spring 2024 LEAP mathematics assessment than those who had negligible, low, or medium usage levels.

Table 12
Mean scores (pretest and posttest), by usage level (4-level)

Level of Usage	Timepoint	Unadjusted Mean (SD)	Adjusted Mean
Negligible	Pretest	729.69 (31.08)	
	Posttest	728.99 (33.45)	730.27
Low Usage	Pretest	719.40 (23.87)	
	Posttest	718.74 (22.13)	727.20
Medium Usage	Pretest	730.14 (26.40)	
	Posttest	730.59 (24.64)	730.93



High Usage	Pretest	743.17 (27.54)	
	Posttest	743.58 (27.88)	734.64

Note. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

The results of the analytic model are shown in Table 13, with the full model shared in Appendix C. These results suggest that high level of Zearn usage was associated with higher LEAP assessment mathematics scores (+4.37 points) than a negligible level of Zearn usage. However, low Zearn usage was associated with lower spring LEAP mathematics scores (-3.07) compared to negligible Zearn usage. Again, due to pre-intervention differences in usage between the two comparison groups, these results cannot be considered causal in nature.

Table 13
Association of level of usage with spring scores (4-level)

Coefficient	Estimate	SE	df	t	p value
Low Usage	-3.07	0.96	4,936.85	-3.21	0.001
Medium Usage	0.66	0.98	4,374.21	0.67	0.502
High Usage	4.37	1.06	3,582.94	4.13	<.001

Note. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lesson and serves as the reference group.

We also conducted analyses with the same covariates but replaced the independent variable with continuous usage variables: total on-grade level lesson completions in the first model and average lessons completed per week in the second model. The results from these analytic models are shown in Table 14, and the complete models are shared in Appendix C. In both models, the greater number of unique on-grade level lessons completed, and the higher average lessons completed per week were associated with higher spring 2024 LEAP mathematics assessment scores, holding prior achievement and demographic variables equal.

Table 14
Association of continuous measures of usage with spring scores

Coefficient	Estimate	SE	df	t	p value
Total Unique On-Grade Lesson Completions	0.04	0.01	3,115.88	3.95	<.001
Average Lessons Completed Per Week	1.69	0.28	3,540.65	6.07	<.001

These findings should be considered exploratory and non-experimental but suggest that greater usage of the Zearn tutoring program may be associated with greater academic achievement.



Comparison Group 2: Impact of Zearn Math Supplemental Use Compared to Zearn Users without Implementation Support

Key Findings

- > Students using Zearn Supplemental scored 6.94 points higher on the LEAP mathematics assessment ($p < .05$) compared to students from typical Zearn user districts, with an effect size of +0.23 *SD*.
- > No differential effects by student characteristics were identified. Students had similar outcomes across race/ethnicity groups, English learner status, economic disadvantage status, and special education participation.
- > Students completing ≥ 90 lessons outscored those with lower usage by 6.12 points ($p < .05$; effect size +0.20 *SD*), regardless of treatment condition.
- > Students with high levels of usage (lessons completed and average lessons completed per week) outscored those with negligible or lower usage levels.

This section describes findings related to Comparison Group 2, comparing the effects of Zearn Math Supplemental against students from districts using Zearn without implementation support. Because Zearn could be implemented in other districts without school-level coaching, we defined these user districts as those that did not participate in the pilot of Zearn Supplemental. Schools from these districts had to use Zearn in a capacity of greater than one lesson per week on average in the year prior to the study.

Prior to matching, 50 comparison schools with 10,321 students and 25 treatment schools with 4,730 students were identified with non-missing spring 2023 LEAP Math scores and demographic and usage data. As described in Tables 15 and 16, school- and student-level differences existed across multiple variables, including prior mathematics achievement, grade, student race/ethnicity, economically disadvantaged status, and special education status. Thus, propensity score matching (PSM) was used to create a comparison group of students that was as similar as possible on observable factors to treatment students (see Tables 15 and 16).



Table 15
Comparison Group 2 school level baseline equivalence, prior to and post-matching, final analytic sample

Category	Level	Pre-Matching Comparison	Pre-Matching Treatment	Post-Matching Comparison	Post-Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
<i>n</i>		50	25	25	25	24	24	
School Attendance Rate Mean (SD)		93.27 (1.34)	92.76 (2.14)	93.23 (1.26)	92.76 (2.14)	93.19 (1.27)	92.76 (2.19)	0.240 (0.41)
% of Students 2022-2023 LEAP Mastery+ Grade 3-8 Mean (SD)		30.94 (16.36)	25.28 (20.31)	27.64 (16.93)	25.28 (20.31)	27.71 (17.29)	26.12 (20.29)	0.084 (0.77)
Total Enrollment Mean (SD)		406.62 (144.19)	369.80 (162.74)	424.04 (167.06)	369.80 (162.74)	405.62 (142.39)	377.38 (161.68)	0.185 (0.52)
LEP Enrollment Rate Mean (SD)		0.04 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.031 (0.92)
Economically Disadvantaged Rate Mean (SD)		0.77 (0.16)	0.80 (0.20)	0.78 (0.17)	0.80 (0.20)	0.79 (0.17)	0.80 (0.20)	0.081 (0.78)
Black/African American Enrollment Rate Mean (SD)		0.43 (0.34)	0.55 (0.28)	0.48 (0.30)	0.55 (0.28)	0.49 (0.30)	0.54 (0.27)	0.150 (0.61)
White Enrollment Rate Mean (SD)		0.41 (0.28)	0.33 (0.30)	0.37 (0.26)	0.33 (0.30)	0.36 (0.26)	0.35 (0.30)	0.040 (0.89)
Latino/a/e Enrollment Rate Mean (SD)		0.09 (0.08)	0.08 (0.07)	0.07 (0.07)	0.08 (0.07)	0.07 (0.07)	0.08 (0.07)	0.141 (0.63)
Grades 4-8 Enrollment Mean (SD)		220.08 (161.19)	210.56 (133.46)	285.60 (190.01)	210.56 (133.46)	261.42 (149.72)	218.04 (130.87)	0.308 (0.29)
Proportion of Active Zearn Users in School (Prior year) Mean (SD)		0.51 (0.28)	0.50 (0.24)	0.63 (0.30)	0.50 (0.24)	0.62 (0.30)	0.52 (0.23)	0.374 (0.20)

Table 16



Comparison Group 2 student level baseline equivalence, prior to and post-matching, final analytic sample

Category	Level	Pre-Matching Comparison	Pre-Matching Treatment	Post-Matching Comparison	Post-Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
n		10321	4730	3571	3571	2436	2576	
2022-2023 LEAP Math Level N (%)	Advanced	318 (3.1)	164 (3.5)	74 (2.1)	139 (3.9)	83 (3.4)	86 (3.3)	
	Mastery	2898 (28.1)	1086 (23.0)	841 (23.6)	796 (22.3)	535 (22.0)	618 (24.0)	
	Basic	3184 (30.8)	1243 (26.3)	950 (26.6)	961 (26.9)	793 (32.6)	770 (29.9)	
	Approaching Basic	2765 (26.8)	1428 (30.2)	1089 (30.5)	1064 (29.8)	777 (31.9)	798 (31.0)	
	Unsatisfactory	1156 (11.2)	809 (17.1)	617 (17.3)	611 (17.1)	248 (10.2)	304 (11.8)	
2022-2023 LEAP Math Score Mean (SD)		735.11 (29.79)	729.89 (32.00)	728.98 (30.28)	730.10 (32.21)	733.06 (28.85)	732.50 (29.44)	0.019 (0.49)
Race/Ethnicity N (%)	Black or African-American	4027 (39.0)	2857 (60.4)	2076 (59.3)	2096 (59.8)	1388 (57.0)	1526 (59.2)	
	Latino/a/e	847 (8.2)	412 (8.7)	207 (5.9)	297 (8.5)	148 (6.1)	216 (8.4)	
	Other	804 (7.8)	145 (3.1)	211 (6.0)	95 (2.7)	157 (6.4)	69 (2.7)	
	White	4643 (45.0)	1316 (27.8)	1009 (28.8)	1015 (29.0)	743 (30.5)	765 (29.7)	
Economically Disadvantaged Status N (%)	Y	7764 (75.2)	3854 (81.5)	2904 (81.3)	2888 (80.9)	1965 (80.7)	2096 (81.4)	
	N	2557 (24.8)	876 (18.5)	667 (18.7)	683 (19.1)	471 (19.3)	480 (18.6)	
English Learner (EL) Status N (%)	Y	266 (2.6)	130 (2.7)	88 (2.5)	87 (2.4)	52 (2.1)	57 (2.2)	
	N	10055 (97.4)	4600 (97.3)	3483 (97.5)	3484 (97.6)	2384 (97.9)	2519 (97.8)	
Special Education Status N (%)	Y	2477 (24)	946 (20)	806 (23)	666 (19)	536 (22)	464 (18)	
	N	10321 (76)	4730 (80)	2697 (77)	2837 (81)	1900 (78)	2112 (82)	



The final Comparison Group 2 analytic sample at baseline, described in the rightmost columns in Tables 15 and 16, consisted of 2,576 treatment students and 2,436 comparison students across 25 treatment and 25 comparison schools. The final column shows the standardized mean differences between treatment and comparison group schools and students. Most importantly, schools and students were similar in their prior academic achievement and in most demographic variables, as indicated by the standardized mean differences being lower than 0.25 SDs. Comparison schools had somewhat higher school enrollment (406 vs. 377 students, SMD = 0.185) and higher prior rates of active Zearn users (0.62 vs. 0.52, SMD = 0.374). But these differences did not appear at the student level.

Research Question 2: Impact Analysis

This first analysis depicts the adjusted and unadjusted descriptive trends in LEAP scores from EOY to EOY for treatment and matched comparison students in the 2023-24 school year. Table 17 depicts the unadjusted and adjusted means for the treatment and comparison groups at baseline and posttest.

Table 17

Mean scores (pretest and posttest), by treatment group

Time	Unadjusted Comparison Mean (SD)	Unadjusted Treatment Mean (SD)	Adjusted Comparison Mean	Adjusted Treatment Mean	Impact Estimate (SE)	p Value	Effect Size
Pretest	733.06 (28.85)	732.50 (29.44)					
Posttest	731.05 (29.62)	732.16 (30.35)	727.92	734.86	6.94 (2.48)	.01	+0.23

This multiple regression analysis controlled for 2022-23 EOY LEAP Math scores and included student and school-level demographic variables. All continuous covariates were grand mean centered to enable interpretation of the intercept. Trends are shown for all students in the analytic sample. The full model can be found in Appendix D.

The Zearn Math Supplemental program was shown to have a statistically significant ($p = .01$) positive impact on LEAP scores. The regression estimate can be interpreted as the average difference in spring 2024 LEAP mathematics scores between treatment and matched comparison students. Thus, treatment students outscored matched comparison students (non-Zearn users) by 6.94 points on the Spring LEAP assessment, controlling for prior achievement and demographic variables. The effect size of +0.23 SDs is indicative of a large practical impact of Zearn on LEAP scores (Kraft, 2020).

Research Question 3: Subgroup Analysis

The next section presents the effects of the Zearn Math Supplemental intervention for different student and subgroups. Table 18 displays the group means for the two-level model, with students within schools, while the full model with interactions is shown in



Appendix D. Differential effects were measured based on race/ethnicity, baseline achievement percentile, economically disadvantaged status, special education status, and English Language Learner status. As with Comparison Group 1, no differential effects were identified by student characteristic. Differential effects were identified by grade with somewhat larger impacts in 8th grade. However given the nature of the scale, we choose to report the adjusted means by condition in the scale of the test for clearer interpretation by practitioners familiar with the LEAP assessment

Table 18

Subgroup analysis results, Zearn impacts on LEAP math scores: Group means

Category	Sub-Category	Comparison			Treatment		
		Estimate	SE	p value	Estimate	SE	p value
Race/Ethnicity	Black or African-American	727.16	1.43	<.001	733.61	1.36	<.001
	Latino/a/e	729.39	2.02	<.001	737.67	1.77	<.001
	Other	730.31	1.88	<.001	734.90	2.29	<.001
	White	728.67	1.56	<.001	736.60	1.50	<.001
Baseline Percentile	25th	703.54	1.51	<.001	709.74	1.43	<.001
	50th	717.58	1.49	<.001	724.51	1.44	<.001
	75th	731.99	1.50	<.001	740.47	1.44	<.001
	100th	760.51	1.56	<.001	767.25	1.47	<.001
Economically Disadvantaged Status	Y	727.11	1.42	<.001	734.29	1.35	<.001
	N	731.40	1.59	<.001	737.30	1.53	<.001
Special Education Status	Y	724.38	1.62	<.001	730.54	1.61	<.001
	N	728.42	1.41	<.001	735.47	1.34	<.001
English Language Learner Status	Y	727.58	2.78	<.001	732.44	2.68	<.001
	N	727.93	1.41	<.001	734.91	1.34	<.001
Grade	5	727.92	1.63	<.001	736.24	1.45	<.001
	6	726.50	1.59	<.001	734.65	1.45	<.001
	7	727.83	1.58	<.001	734.95	1.49	<.001
	8	729.23	1.57	<.001	733.24	1.66	<.001

Note. While 4th grade was included in the model, the sample size for 4th grade was so small that we chose not to include those results.

Research Question 4: Usage Analysis

In the next section, we examine the effect of Zearn on student outcomes for those who received the full treatment, defined as 90 unique on-grade-level lessons over the course of the treatment period. As shown in Table 19, there were 5,012 individual students with Zearn usage data across treatment and comparison conditions in Comparison Group 2.



Table 19
Descriptive statistics on usage by condition

Category	Level	Overall	Comparison	Treatment	<i>p</i>	SMD
<i>n</i>		5012	2436	2576		
Total Unique On-Grade Lesson Completions Mean (SD)		61.31 (40.99)	47.64 (33.82)	74.25 (42.93)	<0.001	0.689
Zearn Usage Level <i>N</i> (%)	Inadequate	3509 (70.0)	2100 (86.2)	1409 (54.7)		
	Adequate	1503 (30.0)	336 (13.8)	1167 (45.3)		

Note. Adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique on-grade-level lessons.

In the analysis of the effect of Zearn on student outcomes for those who received the full treatment, hierarchical linear models (HLM) were used, with students nested within schools as in the previous analyses. However, the treatment variable was replaced with student usage as the independent variable. As in the previous analytic models, student spring 2024 LEAP mathematics scores were the dependent variable. The unadjusted mean scores at pretest and posttest are described in Table 20. As in the previous section, the two groups were found to be significantly different at baseline ($ES = +0.50$). These findings should be considered exploratory and suggestive.

Table 20
Mean scores (pretest and posttest), by usage level (2-level)

Timepoint	Unadjusted Inadequate (<90) Mean (SD)	Unadjusted Adequate (≥ 90) Mean (SD)	Adjusted Inadequate (<90) Mean	Adjusted Adequate (≥ 90) Mean	Impact Estimate (SE)	<i>p</i> value	Effect Size
Pretest	728.39 (28.93)	743.01 (27.04)					+0.50
Posttest	726.74 (29.47)	743.01 (28.09)	729.73	735.85	6.12 (0.65)	<.001	+0.20

Note. Adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique on-grade-level lessons.

The results from the HLM model are described in Table 20, while the full model is shared in Appendix D. The table shows that students receiving the full treatment outscored students who completed fewer than 90 on-grade-level lessons by 6.12 points on the spring LEAP assessment, controlling for prior achievement and demographic variables. The effect size of .20 SDs is indicative of a medium-to-large practical impact



of Zearn on LEAP scores (Kraft, 2020).

Research Question 5: Exploratory Usage Analysis

Finally, we conducted an exploratory analysis with each dataset in which we examined whether the designed full intervention of 90 lessons was indeed the tipping point for impact. To do so, we examined different usage variables as provided by Zearn through the Louisiana Department of Education. Descriptive usage statistics including Comparison Group 2 are described in Table 21.

Treatment students had higher rates of total lessons completed; unique lessons completed; lessons completed above and on grade level; average lessons completed per week; and adequate or high usage.

Table 21
Descriptive statistics on usage by condition

Category	Level	Overall	Comparison	Treatment	<i>p</i>	SMD
<i>n</i>		5012	2436	2576		
Total Unique Below-Grade Level Lesson Completions Mean (SD)		2.93 (10.53)	3.11 (11.01)	2.76 (10.04)	.247	0.033
Total Unique Above-Grade Level Lesson Completions Mean (SD)		1.48 (8.50)	1.11 (7.36)	1.83 (9.44)	.003	0.085
Total Unique On-Grade Level Lesson Completions Mean (SD)		61.31 (40.99)	47.64 (33.82)	74.25 (42.93)	<0.001	0.689
Total Lesson Completions (including repeated lessons) Mean (SD)		72.40 (52.01)	56.17 (41.78)	87.76 (55.93)	<0.001	0.640
Total Unique Lesson Completions Mean (SD)		65.72 (45.25)	51.85 (37.31)	78.84 (48.10)	<0.001	0.627
Average Lessons Completed Per Week Mean (SD)		2.17 (1.35)	1.88 (1.22)	2.45 (1.41)	<0.001	0.425
Zearn Usage Level (2-levels) <i>N</i> (%)	Inadequate	3509 (70.0)	2100 (86.2)	1409 (54.7)		
	Adequate	1503 (30.0)	336 (13.8)	1167 (45.3)		
Zearn Usage Level (4-levels) <i>N</i> (%)	Negligible	1372 (27.4)	862 (35.4)	510 (19.8)		



Low Usage	1115 (22.2)	696 (28.6)	419 (16.3)
Medium Usage	1022 (20.4)	542 (22.2)	480 (18.6)
High Usage	1503 (30.0)	336 (13.8)	1167 (45.3)

Note. In the 2-Level model, adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique on-grade-level lessons. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

As noted in the previous section, these analyses were based on Zearn usage (total unique on-grade-level lessons completed) split into two levels (greater than or less than 90 on-grade-level lessons completed) and four levels (Negligible, Low, Medium, and High). Table 22 describes the pretest and posttest means for students in each of the four levels.

Table 22
Mean scores (pretest and posttest), by usage level (4-level)

Level of Usage	Timepoint	Unadjusted Mean (SD)	Adjusted Mean
Negligible	Pretest	727.25 (32.81)	726.54
	Posttest	723.44 (34.54)	
Low Usage	Pretest	725.93 (25.79)	728.98
	Posttest	724.24 (24.66)	
Medium Usage	Pretest	732.60 (26.04)	733.54
	Posttest	733.91 (25.41)	
High Usage	Pretest	743.01 (27.04)	736.67
	Posttest	743.01 (28.09)	

Note. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

The results of the analytic model are shown in Table 23, with the full model shared in Appendix D. Based on this analysis, high Zearn usage (90 or more on-grade-level lessons completed) was associated with higher spring 2024 LEAP mathematics assessment performance than negligible, low, or medium usage levels. These results suggest that students with high levels of Zearn usage outscored students with negligible usage by 10.14 points on the spring LEAP assessment, controlling for prior achievement and demographic variables. This association was statistically significant and positive.

Table 23
Association of level of usage with spring scores (4-level)



Coefficient	Estimate	SE	df	t value	p value
Low Usage	2.44	0.66	4,904.48	3.70	<.001
Medium Usage	7.01	0.71	4,333.37	9.80	<.001
High Usage	10.14	0.79	4,062.08	12.88	<.001

Note. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

As with the first comparison group, we also conducted analyses using the same covariates but with alternate independent continuous usage variables: In the first model, total on-grade level lesson completions, in the second model, average lessons completed per week. The results from these analytic models are shown in Table 24, and the complete models are shared in Appendix D. In both models, a greater number of unique on-grade level lessons completed, and a higher average number of lessons completed per week were significantly associated with higher spring 2024 LEAP mathematics assessment scores, holding prior achievement and demographic variables equal.

Table 24
Association of continuous measures of usage with spring scores

Coefficient	Estimate	SE	df	t value	p value
Total Unique On-Grade Lesson Completions	0.10	0.01	3,692.33	13.37	<.001
Average Lessons Per Week	3.33	0.22	4,459.69	15.16	<.001

The findings in this section examined the impact of the Zearn Supplemental program compared to Zearn implementation without supports. These findings should be considered exploratory and non-experimental but suggest as did the first comparison that greater usage of the Zearn tutoring program may be associated with greater academic achievement.

In the next section, we will examine Zearn Supplemental districts in contrast to districts that were viewed as similar in implementation by Zearn.

Comparison Group 3: Zearn Math Supplemental Compared to Zearn-Identified Intended User Districts

Key Findings



- Students using Zearn Supplemental scored 2.59 points higher on the LEAP mathematics assessment than students from similar districts (as identified by Zearn), with an effect size of +0.09 *SD*, but this effect was not statistically significant ($p > .05$).
- Differential effects were identified for race/ethnicity. Black, Latino/a/e, and White students had larger effects than students from other race/ethnicity groups. Students had similar outcomes across English learner status, economic disadvantage status, and special education participation.
- Students completing ≥ 90 lessons outscored those with lower usage by 5.77 points ($p < .05$; effect size +0.21 *SD*).
- Students with high levels of usage (lessons completed and average lessons completed per week) outscored those with negligible or lower usage levels.

This section describes findings related to Comparison Group 3, comparing the effects of Zearn Math Supplemental Case use against districts deemed by Zearn as intending to implement Zearn with fidelity.

Prior to matching, 25 comparison schools with 5,827 students from three districts and 25 treatment schools with 4,730 students from the treatment districts were identified with non-missing spring 2023 LEAP Math scores and demographic and usage data. As described in Tables 25 and 26, treatment and comparison groups differed in school-level K-8 enrollment and student-level prior achievement, economic disadvantaged status, and special education status. Thus, as in the previous analytic groups, propensity score matching (PSM) was used to create a comparison group of students that was as similar as possible to treatment students (see Tables 25 and 26).

As in the previous analytic groups, the *matchMulti* package (Keele et al., 2024) in *R* (v 4.4.3; R Core Team, 2024), multilevel matching for data with cluster level treatments and individual-level outcomes using a network optimization algorithm was used to create the sample.

The final Comparison Group 3 analytic sample at baseline, described in the rightmost columns in Tables 25 and 26, consisted of 2,256 treatment students and 2,306 comparison students across 24 treatment and 24 comparison schools. The final column shows the standardized mean differences between treatment and comparison group schools and students. Most importantly, though there were differences at the school level in the proportion of students who reached LEAP Mastery or higher in prior achievement (SMD = 0.384), students were similar in their prior academic achievement and in most demographic variables, as indicated by the standardized mean differences being lower than 0.25 *SDs*. Comparison schools had somewhat higher school enrollment (493 vs. 409 students, SMD = 0.470), and lower rates of enrolled economically disadvantaged students (SMD = 0.35), Black or African American students (SMD = 0.417), and higher White enrollment (SMD = 0.283). But these differences did not appear at the student level.



Table 25
Comparison Group 3 school level baseline equivalence, prior to and post-matching, final analytic sample

Category	Level	Pre-Matching Comparison	Pre-Matching Treatment	Post-Matching Comparison	Post-Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
<i>n</i>		25	25	25	25	24	24	
School Attendance Rate Mean (SD)		93.17 (1.50)	92.76 (2.14)	93.17 (1.50)	92.76 (2.14)	93.18 (1.53)	92.76 (2.19)	0.225 (0.44)
% of Students 2022-2023 LEAP Mastery+ Grade 3-8 Mean (SD)		32.40 (18.76)	25.28 (20.31)	32.40 (18.76)	25.28 (20.31)	33.54 (18.25)	26.12 (20.29)	0.384 (0.19)
Total Enrollment Mean (SD)		441.80 (166.32)	369.80 (162.74)	441.80 (166.32)	369.80 (162.74)	446.42 (168.26)	377.38 (161.68)	0.418 (0.15)
LEP Enrollment Rate Mean (SD)		0.04 (0.04)	0.03 (0.04)	0.04 (0.04)	0.03 (0.04)	0.04 (0.04)	0.03 (0.04)	0.167 (0.57)
Economically Disadvantaged Rate Mean (SD)		0.74 (0.18)	0.80 (0.20)	0.74 (0.18)	0.80 (0.20)	0.73 (0.18)	0.80 (0.20)	0.350 (0.23)
Black/African American Enrollment Rate Mean (SD)		0.42 (0.39)	0.55 (0.28)	0.42 (0.39)	0.55 (0.28)	0.40 (0.38)	0.54 (0.27)	0.417 (0.16)
White Enrollment Rate Mean (SD)		0.41 (0.29)	0.33 (0.30)	0.41 (0.29)	0.33 (0.30)	0.43 (0.28)	0.35 (0.30)	0.283 (0.33)
Latino/a/e Enrollment Rate Mean (SD)		0.09 (0.08)	0.08 (0.07)	0.09 (0.08)	0.08 (0.07)	0.10 (0.08)	0.08 (0.07)	0.178 (0.54)
Grades 4-8 Enrollment Mean (SD)		255.44 (176.98)	210.56 (133.46)	255.44 (176.98)	210.56 (133.46)	252.29 (180.07)	218.04 (130.87)	0.218 (0.46)
Proportion of Active Zearn Users in School (Prior Year) Mean (SD)		0.58 (0.30)	0.50 (0.24)	0.58 (0.30)	0.50 (0.24)	0.57 (0.30)	0.52 (0.23)	0.182 (0.53)



Table 26
Comparison Group 3 student level baseline equivalence, prior to and post-matching, final analytic sample

Category	Level	Pre-Matching Comparison	Pre-Matching Treatment	Post-Matching Comparison	Post-Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
<i>n</i>		5837	4730	2878	2878	2306	2256	
2022-2023 LEAP Math Level <i>N (%)</i>	Advanced	194 (3.3)	164 (3.5)	97 (3.4)	86 (3.0)	84 (3.6)	82 (3.6)	
	Mastery	1606 (27.5)	1086 (23.0)	697 (24.2)	661 (23.0)	624 (27.1)	663 (29.4)	
	Basic	1733 (29.7)	1243 (26.3)	797 (27.7)	761 (26.4)	677 (29.4)	622 (27.6)	
	Approaching Basic	1554 (26.6)	1428 (30.2)	804 (27.9)	867 (30.1)	594 (25.8)	567 (25.1)	
	Unsatisfactory	750 (12.8)	809 (17.1)	483 (16.8)	503 (17.5)	327 (14.2)	322 (14.3)	
2022-2023 LEAP Math Score Mean (<i>SD</i>)		734.44 (30.90)	729.89 (32.00)	731.27 (31.97)	729.17 (31.46)	733.69 (31.63)	732.29 (31.37)	0.044 (0.13)
Race/Ethnicity <i>N (%)</i>	American Indian or Alaskan	81 (1.4)	9 (0.2)	24 (0.8)	6 (0.2)	20 (0.9)	4 (0.2)	
	Asian	76 (1.3)	34 (0.7)	27 (0.9)	21 (0.7)	26 (1.1)	17 (0.8)	
	Black or African-American	2280 (39.1)	2857 (60.4)	1622 (56.4)	1639 (56.9)	1237 (53.6)	1236 (54.8)	
	Latino/a/e	531 (9.1)	412 (8.7)	199 (6.9)	205 (7.1)	157 (6.8)	165 (7.3)	
Economically Disadvantaged Status <i>N (%)</i>	Y	4186 (71.7)	3854 (81.5)	2221 (77.2)	2347 (81.5)	1738 (75.4)	1784 (79.1)	
	N	1651 (28.3)	876 (18.5)	657 (22.8)	531 (18.5)	568 (24.6)	472 (20.9)	
English Learner (EL) Status Mean <i>N (%)</i>	Y	160 (2.7)	130 (2.7)	72 (2.5)	59 (2.1)	59 (2.6)	38 (1.7)	
	N	5677 (97.3)	4600 (97.3)	2806 (97.5)	2819 (97.9)	2247 (97.4)	2218 (98.3)	
Special Education Status	Y	1518 (26)	946 (20)	720 (25)	547 (19)	576 (25)	451 (20)	



Category	Level	Pre-Matching Comparison	Pre-Matching Treatment	Post-Matching Comparison	Post-Matching Treatment	Final Sample Comparison	Final Sample Treatment	SMD (p)
Mean N (%)	N	4319 (74)	3784 (80)	2158 (75)	2331 (81)	1730 (75)	1805 (80)	



Research Question 2: Impact Analysis

This first analysis depicts the adjusted and unadjusted descriptive trends in LEAP scores from EOY to EOY for treatment and matched comparison students in the 2023-24 school year. Table 27 depicts the unadjusted and adjusted means for the treatment and comparison groups at baseline and posttest.

Table 27

Impact of Zearn Supplemental on achievement and mean scores (pretest and posttest), by treatment group

Time	Unadjusted Comparison Mean (SD)	Unadjusted Treatment Mean (SD)	Adjusted Comparison Mean	Adjusted Treatment Mean	Impact Estimate (SE)	p Value	Effect Size
Pretest	733.69 (31.63)	732.29 (31.37)					
Posttest	732.75 (29.79)	732.83 (31.70)	731.23	733.82	2.59 (2.24)	0.253	+0.09

As in the previous analyses, this multiple regression analysis controlled for 2022-23 EOY LEAP Math scores and included student and school-level demographic variables. All continuous covariates were grand mean centered to enable interpretation of the intercept. Trends are shown for all students in the analytic sample. The full model can be found in Appendix M.

In this analysis, the Zearn Math Supplemental program had a positive but not statistically significant impact on student spring mathematics achievement in relation to those who participated in similar implementation districts ($p = .253$).

Research Question 3: Subgroup Analysis

The next section presents the effects of the Zearn Math Supplemental Use Case intervention for different student subgroups. Table 28 displays the group means for the two-level model, with students within schools, while the full model with interactions is shown in Appendix E. Differential effects were identified along race/ethnicity groups. Black, Latino/a/e, and White students had larger effects than students from other race/ethnicity groups. Students had similar outcomes across English learner status, economic disadvantage status, and special education participation. Differential effects were identified by grade with somewhat larger impacts in 8th grade. However given the nature of the scale, we choose to report the adjusted means by condition in the scale of the test for clearer interpretation by practitioners familiar with the LEAP assessment



Table 28
Subgroup analysis results, Zearn impacts on LEAP math scores: Group means

Category	Sub-Category	Comparison			Treatment		
		Estimate	SE	p value	Estimate	SE	p value
Race/Ethnicity	Black or African-American	729.48	1.25	<.001	732.39	1.31	<.001
	Latino/a/e	730.81	1.92	<.001	736.97	1.83	<.001
	Other	735.80	1.88	<.001	733.09	2.15	<.001
	White	733.52	1.42	<.001	735.53	1.51	<.001
Baseline Percentile	25th	703.19	1.37	<.001	707.33	1.44	<.001
	50th	721.05	1.35	<.001	723.05	1.42	<.001
	75th	737.17	1.37	<.001	741.95	1.44	<.001
	100th	763.09	1.44	<.001	770.93	1.50	<.001
Economically Disadvantaged Status	Y	730.86	1.24	<.001	733.10	1.32	<.001
	N	732.50	1.38	<.001	736.30	1.49	<.001
Special Education Status	Y	726.61	1.46	<.001	730.04	1.59	<.001
	N	731.89	1.23	<.001	734.36	1.30	<.001
English Language Learner Status	Y	733.19	2.62	<.001	729.24	2.96	<.001
	N	731.19	1.23	<.001	733.93	1.30	<.001
Grade	5	731.13	1.38	<.001	734.27	1.41	<.001
	6	732.21	1.36	<.001	734.23	1.43	<.001
	7	733.73	1.46	<.001	734.15	1.45	<.001
	8	725.90	1.93	<.001	731.46	1.60	<.001

Note. While 4th grade was included in the model, the sample size for 4th grade was so small that we chose not to include those results.

Research Question 4: Usage Analysis

In the next section, we examined the effect of Zearn on student outcomes for those who received the full treatment, defined as 90 unique on-grade-level lessons over the course of the treatment period. As shown in Table 29, there were 5,060 individual students with Zearn usage data across treatment and comparison conditions in Comparison Group 3. As would be expected, given the opt-in process to participate in Zearn Math Supplemental and district- and school-level matching, students in treatment districts completed a greater number of unique lessons and were more likely to have completed an adequate number of levels.



Table 29
Descriptive statistics on usage by condition

Category	Sub-Category	Overall	Comparison	Treatment	<i>p</i>	SMD
<i>n</i>		4562	2306	2256		
Total Unique On-Grade Lesson Completions Mean (SD)		61.63 (40.64)	53.19 (35.15)	70.27 (43.93)	<0.001	0.429
Zearn Usage Level <i>N</i> (%)	Inadequate	3247 (71.2)	1906 (82.7)	1341 (59.4)	<0.001	0.529
	Adequate	1315 (28.8)	400 (17.3)	915 (40.6)		

Note. Adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique on-grade-level lessons.

In the analysis of the effect of Zearn on student outcomes for those who received the full treatment, hierarchical linear models (HLM) were used, with students nested within schools as in the previous analyses. However, the treatment variable was replaced with student usage as the independent variable. As in the previous analytic models, student spring 2024 LEAP mathematics scores were the dependent variable. The unadjusted and adjusted mean scores at pretest and posttest are described in Table 30. As in the previous section, due to the differences in prior Zearn usage between the two groups ($ES = +0.66$), these findings should be considered exploratory and suggestive.

Table 30
Mean scores (pretest and posttest), by usage level (2-level)

Timepoint	Unadjusted Inadequate (<90) Mean (SD)	Unadjusted Adequate (≥ 90) Mean (SD)	Adjusted Inadequate (<90) Mean	Adjusted Adequate (≥ 90) Mean	Impact Estimate (SE)	<i>p</i> Value	Effect Size
Pretest	727.39 (31.25)	746.86 (27.61)					+0.66
Posttest	727.01 (30.24)	747.04 (27.14)	730.94	736.71	5.77 (0.64)	<0.001	0.212

Note. Adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique on-grade-level lessons.

The full HLM model is shared in Appendix E. The table shows that students receiving the full treatment outscored students who completed fewer than 90 on-grade-level lessons by 4.74 points on the spring LEAP assessment, controlling for prior achievement and demographic variables. The effect size of 0.21 *SDs* is indicative of a large practical impact of Zearn on LEAP scores (Kraft, 2020).



Research Question 5: Exploratory Usage Analysis

Finally, we conducted an exploratory analysis with each dataset in which we examined whether the designed full intervention of 90 lessons was indeed the tipping point for impact. To do so, we examined different usage variables as provided by Zearn through the Louisiana Department of Education. Descriptive usage statistics for Comparison Group 3 are described in Table 31.

Zearn provided variables counting the total number of lessons completed; the total number of unique lessons completed below grade level, on grade level, and above grade level; and weeks of Zearn usage. Using these variables, we also calculated the total number of unique lessons completed (adding together their below, on grade level, and above grade level unique lessons) and the average number of lessons per week (dividing total lessons by weeks of Zearn usage). In addition, we created a four-level categorical variable using cutoffs from Zearn, and a two-level categorical variable based on the 90 on-grade-level lessons threshold. In the 4-level model, high usage is defined as completing 90 on-grade-level unique lessons or more; medium usage is defined as completing between 60 and 90 unique on-grade-level lessons; low usage is defined as completing between 30 and 60 unique on-grade-level lessons; and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

As shown in Table 31, students in the treatment group districts completed more total lessons, more unique lessons, and completed more lessons per week.



Table 31
Descriptive statistics on usage by condition

Category	Level	Overall	Comparison	Treatment	<i>p</i>	SMD
<i>n</i>		4562	2306	2256		
Total Unique Below-Grade Level Lesson Completions Mean (SD)		3.40 (12.66)	4.62 (15.81)	2.15 (8.11)	<0.001	0.196
Total Unique Above-Grade Level Lesson Completions Mean (SD)		1.60 (8.22)	1.47 (8.52)	1.73 (7.90)	.278	0.032
Total Unique On-Grade Level Lesson Completions Mean (SD)		61.63 (40.64)	53.19 (35.15)	70.27 (43.93)	<0.001	0.429
Total Lesson Completions (including repeated lessons) Mean (SD)		73.63 (51.56)	64.17 (44.19)	83.30 (56.52)	<0.001	0.377
Total Unique Lesson Completions Mean (SD)		66.63 (44.09)	59.27 (37.92)	74.15 (48.47)	<0.001	0.342
Average Lessons Completed Per Week Mean (SD)		2.19 (1.32)	2.08 (1.20)	2.30 (1.43)	<0.001	0.170
Zearn Usage Level (2-levels) <i>N</i> (%)	Inadequate	3247 (71.2)	1906 (82.7)	1341 (59.4)		
	Adequate	1315 (28.8)	400 (17.3)	915 (40.6)		
	Negligible	1192 (26.1)	676 (29.3)	516 (22.9)		
Zearn Usage Level (4-levels) <i>N</i> (%)	Low Usage	1054 (23.1)	655 (28.4)	399 (17.7)		
	Medium Usage	1001 (21.9)	575 (24.9)	426 (18.9)		
	High Usage	1315 (28.8)	400 (17.3)	915 (40.6)		

Note. In the 2-Level model, adequate usage is defined as completing 90 unique on-grade-level lessons or more over the course of the intervention period. Inadequate usage is defined as completing fewer than 90 unique lessons. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

The models here were the same as in the previous two sections. As can be seen in Table 32, which describes the mean pretest and posttest scores for students in each of



the four usage categories, students with high usage performed higher on the Spring 2024 LEAP mathematics assessment than those who had negligible, low, or medium usage levels.

Table 32
Mean scores (pretest and posttest), by usage level (4-level)

Level of Usage	Timepoint	Unadjusted Mean (SD)	Adjusted Mean
Negligible	Pretest	724.73 (36.48)	729.06
	Posttest	723.42 (36.90)	
Low Usage	Pretest	724.39 (27.89)	730.52
	Posttest	724.23 (24.93)	
Medium Usage	Pretest	733.70 (26.63)	733.02
	Posttest	734.22 (24.82)	
High Usage	Pretest	746.86 (27.61)	737.22
	Posttest	747.04 (27.14)	

Note. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

The results of the analytic model are shown in Table 33, with the full model shared in Appendix E. These results suggest that students with high levels of Zearn usage were associated with an additional 8.16 points on the LEAP assessment compared to those with negligible usage levels, when controlling for prior achievement and demographic variables; students with medium usage levels were associated with an additional 3.96 points; and students with low usage were associated with an additional 1.46 points. These results were statistically significant and positive.

Table 33
Association of level of usage with spring scores (4-level)

Coefficient	Estimate	SE	df	t value	p value
Low Usage	1.46	0.67	4,515.86	2.17	.030
Medium Usage	3.96	0.72	4,344.46	5.49	<.001
High Usage	8.16	0.80	3,657.63	10.17	<.001

Note. In the 4-level model, high usage is defined as completing 90 unique on-grade-level lessons or more, medium usage is defined as completing between 60 and 90 unique on-grade-level lessons, low usage is defined as completing between 30 and 60 unique on-grade-level lessons, and negligible usage is defined as completing fewer than 30 unique on-grade-level lessons.

We also conducted the analysis using total on-grade-level lesson completions as a continuous variable and average lessons per week as a continuous variable. The results from these analytic models are shown in Table 34, and the complete models are shared in Appendix E. In both models, students with a greater number of unique on-grade lessons completed and higher average lessons per week scored higher on the Spring 2024 LEAP mathematics assessment, holding prior achievement and demographic variables equal.



Table 34
Association of continuous measures of usage with spring scores

Coefficient	Estimate	SE	df	t value	p value
Total Unique On-Grade Lesson Completions	0.08	0.01	3,431.23	10.21	<.001
Average Lessons Completed Per Week	2.61	0.22	4,002.33	11.84	<.001

These findings should be considered exploratory and non-experimental but again suggest that greater usage of the Zearn tutoring program may be associated with greater academic achievement.



DISCUSSION

The purpose of the present study was to evaluate the efficacy of the Zearn Supplemental with dedicated implementation support program in the State of Louisiana. The main outcome measures included LEAP mathematics scores across three related comparison groups. First, Zearn Supplemental students were compared to non-Zearn students (defined as those from districts with less than one lesson completed per week on average). Second, Zearn Supplemental students were compared to students from districts using Zearn without implementation support. And finally, Zearn Supplemental students were compared to students from districts identified by Zearn as implementing Zearn in a similar fashion and fidelity to the treatment supplemental districts. Zearn usage data were also analyzed, as associations between program usage (lessons completed) and achievement were examined in this study. Findings from impact analyses and supplemental correlational analyses relating to usage were presented in this report. While this study cannot isolate the impact specific to the supplemental Zearn supports, it is interesting to consider how the results of the first comparison group (non-users, impact = 5.97) and the second comparison group (users without implementation supports, impact = 6.94) differed. While we may have hypothesized that the impact would be larger when compared to non-users rather than users because one would expect the support in Zearn Supplemental to increase dosage and thereby increase student outcomes. However, we see the opposite pattern—impacts are higher when comparing Zearn Supplemental to users without implementation supports than non-users. These are not substantially different impacts but do illustrate that further work is needed with prospective assignment of schools to either Zearn without implementation support, Zearn Supplemental, and non-use (control), which would remove concerns about unobserved factors and selection bias which may also be driving these differences.

Limitations

While this study provides evidence of associations between participation in the tutoring program and improved student mathematics achievement, several important limitations must be acknowledged, related to the internal validity of the findings. The study employed a multi-level propensity score matching (PSM) design to construct comparison groups that were similar to treatment students on a range of observed characteristics. One-to-one multi-level matching was conducted in two rounds—first at the district and school levels to create three sets of comparison groups, and subsequently across those groups at the school and student levels. This resulted in three distinct comparison groups. However, none of these groups included a traditional control group. As Zearn was available across the State of Louisiana, we had to create a definition of non-users based on previous usage data: average lessons completed per week at the school level.

Despite efforts to create balance across observed covariates, several potential threats to internal validity remain. First, the non-randomized nature of the study limits causal inference. The validity of propensity score matching relies on the assumption of



selection on observable characteristics or variables, meaning that all confounding variables are measured sufficiently and included in the analysis model (Rubin, 2007). In practice, there are often unmeasured factors (such as parent engagement or school leadership) that may influence outcomes and are not within our ability to measure or include in an analysis. While a randomized control trial would control for these unobserved factors through the act of randomization to treatment or control condition, these trials are often not feasible. However, the potential presence of any such unobserved confounding does introduce bias into the estimated treatment effects.

In addition, although the multi-level matching process used in this study improves comparability across units (here we use districts, schools, students), the complexity of the design introduces additional opportunities for confounding. While we attempted to account for this by creating multiple sets of comparison groups, systematic differences in how districts or schools chose to engage with or implement the tutoring program, for instance, may correlate with student achievement in ways that are not fully captured in the matching covariates. In addition, the comparison group of districts identified by the implementer as “similar” to those districts where the treatment took place may create similar analysis subjects but also introduces the potential for subjective bias in the selection process, which may not align with the formal matching criteria.

Despite these limitations, this study is in line with the study design guidelines set out by What Works Clearinghouse (What Works Clearinghouse, 2022), which notes that statistical adjustment and matching are the two acceptable methods for creating baseline equivalence in quasi-experimental design research. Propensity score matching study designs are widely used and accepted in education research and accepted as a valid quasi-experimental design in the absence of randomization (Evans et al., 2025; Powell et al., 2020; Stuart, 2007). These methods are used to “equate groups on a pretest measure of the outcome variable” and create analysis groups that appear similar on a pretest of the outcome measure and across other characteristics (Powell et al., 2020; What Works Clearinghouse, 2022). In addition, propensity scores have been identified as a robust method for creating matched groups (Stuart & Rubin, 2007). Given this, the treatment effect estimates presented in this study may be interpreted as causal evidence with reservations.

Conclusion

Results of this evaluation showed a statistically significant positive impact of Zearn Supplemental on student LEAP mathematics achievement in comparison to non-Zearn users and in comparison to those using Zearn without implementation supports. However, no statistically significant impact was identified when comparing Zearn Supplemental students to students from districts identified by Zearn as implementing Zearn similarly to the treatment districts. These findings give evidence supporting the efficacy of Zearn Supplemental in later elementary and middle school grades (Grades 4-8). Subgroup analyses suggested that the intervention was effective across subgroups in an equitable manner, with no differential effects found by grade, race/ethnicity, prior achievement percentile, economic disadvantage status, special education status, or English Language Learner status. In addition, analysis suggested



that increased usage of Zearn was associated with increased academic achievement in each comparison group. It is important to consider that this intervention took place across the State of Louisiana, where all districts had the option of using Zearn, though only the three treatment districts received the Zearn Supplemental with dedicated implementation support intervention.



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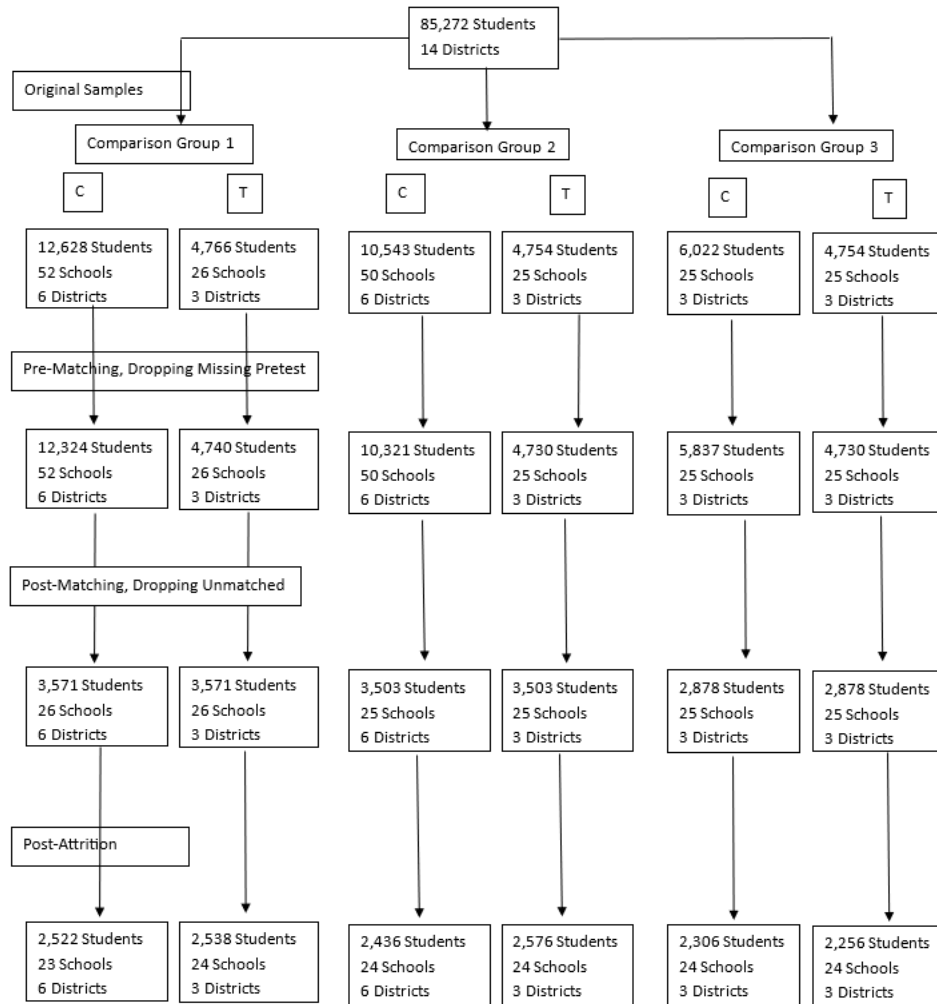


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APPENDIX A: Consort Diagram





APPENDIX B: Attrition Analysis

Table B1

Summary of cluster attrition for outcome variable (CG1 Sample)

C School N	T School N	N School to C	N School to T	Attrited C School	Attrited T School	Overall School Attrition Rate (%)	Differential School Attrition Rate (%)
52	26	26	26	3	2	9.62%	3.85%

Table B2

Summary of student attrition for outcome variable (CG1 Sample)

C Student N	T Student N	N Matched to C	N Matched to T	Attrited C Students	Attrited T Students	Overall Student Attrition Rate (%)	Differential Student Attrition Rate (%)
12,628	4,766	3,571	3,571	1,049	1,033	29.15%	0.45%

Table B3

Summary of cluster attrition for outcome variable (CG2 Sample)

C School N	T School N	N School to C	N School to T	Attrited C School	Attrited T School	Overall School Attrition Rate (%)	Differential School Attrition Rate (%)
50	25	25	25	1	1	4.00%	0.00%

Table B4

Summary of student attrition for outcome variable (CG2 Sample)

C Student N	T Student N	N Matched to C	N Matched to T	Attrited C Students	Attrited T Students	Overall Student Attrition Rate (%)	Differential Student Attrition Rate (%)
10,543	4,754	3,503	3,503	1,067	927	28.46%	4.00%

Table B5

Summary of cluster attrition for outcome variable (CG3 Sample)

C	T	N School	N School	Attrited C	Attrited T	Overall	Differentia
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School <i>N</i>	School <i>N</i>	to C	to T	School	School	School Attrition Rate (%)	I School Attrition Rate (%)
25	25	25	25	1	1	4.00%	0.00%

Table B6

Summary of student attrition for outcome variable (CG3 Sample)

C Student <i>N</i>	T Student <i>N</i>	<i>N</i> Matched to C	<i>N</i> Matched to T	Attrited C Students	Attrited T Students	Overall Student Attrition Rate (%)	Differential Student Attrition Rate (%)
6,022	4,754	2,878	2,878	572	622	20.74%	1.74%



APPENDIX C: Comparison Group 1 Analytic Models

Table C1

Comparison Group 1: Impact analysis of Zearn Supplemental on spring 2024 LEAP Math overall scale scores: Full model

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	727.82	1.57	30.13	463.74	<.001
Treatment	5.97	2.78	30.34	2.15	0.040
Black or African American	-3.66	1.40	5,018.81	-2.62	0.009
White	-0.82	1.44	5,026.91	-0.57	0.572
Latino/a/e	0.56	1.86	5,020.87	0.30	0.765
Grade 4	31.24	6.54	5,011.24	4.78	<.001
Grade 5	3.30	1.25	1,895.86	2.65	0.008
Grade 6	3.70	1.07	3,693.46	3.46	0.001
Grade 7	4.99	0.95	4,767.34	5.25	<.001
English Learner	0.45	2.31	5,005.64	0.19	0.847
Economically Disadvantaged Status	-3.13	0.92	5,027.96	-3.38	0.001
Special Education Status	-3.93	0.97	5,014.91	-4.05	<.001
2022-2023 LEAP Math Score	0.84	0.01	5,026.32	86.09	<.001
Proportion of Black or African American School Enrollment	2.92	4.07	34.67	0.72	0.478
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.19	0.07	32.59	2.84	0.008
District 1	-3.76	3.03	32.52	-1.24	0.224
District 3	-2.12	4.88	29.66	-0.43	0.668
District 31	-4.72	2.75	31.06	-1.71	0.096
District 32	-1.04	3.85	34.55	-0.27	0.789
District 35	-2.79	2.34	31.83	-1.19	0.242
District 37	4.89	2.83	32.97	1.73	0.093
District 49	3.87	3.09	30.77	1.25	0.221
Treatment*Black	2.00	2.34	5,026.69	0.85	0.393
Treatment*White	1.78	2.39	5,026.69	0.74	0.458
Treatment* Latino/a/e	2.20	2.88	5,019.83	0.76	0.445
Treatment*Grade 4	-14.50	9.68	5,013.17	-1.50	0.134
Treatment*Grade 5	-6.67	1.71	1,937.73	-3.91	<.001
Treatment*Grade 6	-7.35	1.50	3,455.30	-4.89	<.001
Treatment*Grade 7	-7.58	1.42	4,089.30	-5.35	<.001
Treatment*English Learner	-1.57	3.28	5,006.50	-0.48	0.633
Treatment*Economically Disadvantaged	1.27	1.35	5,021.85	0.94	0.347
Treatment*Special Education	-0.11	1.39	5,012.30	-0.08	0.934

**Table C2**

Comparison Group 1: Analysis of adequate usage (2-level) on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	729.98	0.62	38.16	1,185.10	<.001
Usage (90 Lessons)	4.74	0.86	4,372.13	5.49	<.001
2022-2023 LEAP Math Score	0.83	0.01	5,035.60	83.65	<.001
Black or African American	-3.11	1.12	5,036.10	-2.77	0.006
White	-0.36	1.14	5,036.16	-0.31	0.755
Latino/a/e	1.47	1.38	5,031.84	1.06	0.288
English Learner	-0.69	1.64	5,020.42	-0.42	0.676
Economically Disadvantaged Status	-2.60	0.68	5,033.21	-3.83	<.001
Special Education Status	-4.18	0.71	5,024.73	-5.85	<.001
Black/African American Enrollment Rate	3.64	3.69	38.18	0.98	0.331
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.22	0.06	36.09	3.57	0.001
District 1	-4.60	2.74	34.86	-1.68	0.102
District 3	-1.83	4.38	31.21	-0.42	0.679
District 31	-3.05	2.72	40.45	-1.12	0.269
District 32	-1.38	3.50	38.20	-0.39	0.696
District 35	2.60	2.35	36.12	1.11	0.275
District 37	3.66	2.55	35.56	1.43	0.161
District 48	4.35	2.52	33.38	1.73	0.093
District 49	3.64	2.78	32.78	1.31	0.200
Grade 4	24.40	4.82	5,024.44	5.06	<.001
Grade 5	-0.37	0.85	1,891.74	-0.43	0.665
Grade 6	-0.36	0.75	3,388.32	-0.48	0.632
Grade 7	1.48	0.70	4,045.80	2.12	0.034

**Table C3**

Comparison Group 1: Analysis of usage (4-level) level on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	730.27	0.69	55.28	1,062.83	<.001
Low Zearn Usage	-3.07	0.96	4,936.85	-3.21	0.001
Medium Zearn Usage	0.66	0.98	4,374.21	0.67	0.502
High Zearn Usage	4.37	1.06	3,582.94	4.13	<.001
2022-2023 LEAP Math Score	0.83	0.01	5,026.32	82.80	<.001
Black or African American	-3.17	1.12	5,034.12	-2.83	0.005
White	-0.36	1.14	5,034.38	-0.32	0.752
Latino/a/e	1.41	1.38	5,029.51	1.02	0.307
English Learner	-0.70	1.64	5,018.13	-0.42	0.672
Economically Disadvantaged Status	-2.60	0.68	5,032.49	-3.84	<.001
Special Education Status	-4.33	0.72	5,022.54	-6.05	<.001
Black/African American Enrollment Rate	3.60	3.73	38.31	0.96	0.341
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.22	0.06	36.57	3.49	0.001
District 1	-4.51	2.77	35.15	-1.63	0.112
District 3	-1.74	4.42	31.30	-0.39	0.696
District 31	-2.74	2.82	44.45	-0.97	0.336
District 32	-1.34	3.53	38.18	-0.38	0.705
District 35	3.03	2.41	38.68	1.26	0.216
District 37	3.70	2.58	35.60	1.44	0.160
District 48	5.07	2.62	37.62	1.93	0.061
District 49	3.65	2.81	32.85	1.30	0.203
Grade 4	24.50	4.82	5,021.74	5.08	<.001
Grade 5	-0.29	0.85	1,960.63	-0.35	0.728
Grade 6	-0.35	0.75	3,553.61	-0.46	0.643
Grade 7	1.59	0.70	4,164.86	2.25	0.024

**Table C4**

*Comparison Group 1: Analysis of total on-grade completed lessons on spring 2024
LEAP Math overall scale scores*

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	729.47	0.69	58.56	1,055.23	<.001
Total Completed On-Grade Lessons	0.04	0.01	3,115.88	3.95	<.001
2022-2023 LEAP Math Score	0.83	0.01	5,022.42	82.96	<.001
Black or African American	-3.10	1.12	5,036.31	-2.76	0.006
White	-0.34	1.15	5,036.04	-0.30	0.764
Latino/a/e	1.51	1.38	5,032.13	1.09	0.275
English Learner	-0.55	1.65	5,020.65	-0.34	0.737
Economically Disadvantaged Status	-2.66	0.68	5,033.97	-3.91	<.001
Special Education Status	-4.08	0.72	5,024.80	-5.69	<.001
Black/African American Enrollment Rate	3.55	3.69	38.40	0.96	0.342
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.22	0.06	36.88	3.54	0.001
District 1	-4.98	2.74	35.23	-1.82	0.077
District 3	-1.87	4.36	31.25	-0.43	0.671
District 31	-3.28	2.82	46.72	-1.16	0.252
District 32	-1.37	3.48	38.31	-0.39	0.697
District 35	1.64	2.38	38.55	0.69	0.495
District 37	3.60	2.54	35.64	1.41	0.166
District 48	3.11	2.58	37.10	1.21	0.235
District 49	3.57	2.77	32.83	1.29	0.207
Grade 4	23.98	4.83	5,024.07	4.96	<.001
Grade 5	-0.41	0.85	1,973.15	-0.48	0.630
Grade 6	-0.48	0.76	3,565.19	-0.64	0.524
Grade 7	1.25	0.70	4,111.70	1.78	0.076

**Table C5**

Comparison Group 1: Analysis of average lessons completed per week on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	729.00	0.71	62.59	1,021.24	<.001
Average Lessons Completed Per Week	1.69	0.28	3,540.65	6.07	<.001
2022-2023 LEAP Math Score	0.83	0.01	4,571.41	79.38	<.001
Black or African American	-2.87	1.13	4,584.10	-2.54	0.011
White	-0.12	1.16	4,585.75	-0.10	0.917
Latino/a/e	1.36	1.40	4,579.46	0.97	0.331
English Learner	0.15	1.68	4,567.93	0.09	0.928
Economically Disadvantaged Status	-1.99	0.71	4,583.70	-2.80	0.005
Special Education Status	-4.04	0.74	4,573.22	-5.45	<.001
Black/African American Enrollment Rate	3.95	3.73	37.51	1.06	0.296
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.24	0.06	35.89	3.87	<.001
District 1	-5.91	2.81	36.18	-2.10	0.042
District 3	-2.00	6.32	127.29	-0.32	0.752
District 31	-5.08	2.82	43.42	-1.80	0.078
District 32	-1.83	3.52	37.10	-0.52	0.606
District 35	0.31	2.41	37.45	0.13	0.899
District 37	3.09	2.57	34.58	1.20	0.238
District 48	1.99	2.59	35.28	0.77	0.447
District 49	5.18	2.85	33.82	1.82	0.078
Grade 4	19.83	5.29	4,574.13	3.75	<.001
Grade 5	-1.65	0.89	1,656.58	-1.85	0.065
Grade 6	-2.20	0.80	3,022.33	-2.75	0.006
Grade 7	0.05	0.73	3,747.84	0.06	0.950



APPENDIX D: Comparison Group 2 Analytic Models

Table D1

Comparison Group 2: Impact analysis of Zearn Supplemental on spring 2024 LEAP Math overall scale scores: Full model

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	727.92	1.41	31.99	517.97	0.000
Treatment	6.94	2.48	32.87	2.80	0.009
Black or African American	-3.15	1.37	4,979.85	-2.30	0.021
White	-1.64	1.42	4,974.93	-1.15	0.249
Latino/a/e	-0.92	1.88	4,979.98	-0.49	0.623
Grade 4	19.80	5.40	4,734.41	3.66	0.000
Grade 5	-1.31	1.45	525.94	-0.90	0.366
Grade 6	-2.73	1.38	590.24	-1.98	0.048
Grade 7	-1.40	1.26	951.60	-1.11	0.268
English Language Learner Status	-0.35	2.46	4,960.72	-0.14	0.887
Economically Disadvantaged Status	-4.29	0.93	4,969.26	-4.63	0.000
Special Education Status	-4.05	0.94	4,955.47	-4.32	0.000
2022-2023 LEAP Math Score	0.81	0.01	4,973.91	84.94	0.000
Black/African American Enrollment Rate	-2.64	3.80	38.90	-0.69	0.492
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.08	0.06	35.94	1.33	0.193
District 8	-1.33	3.42	32.04	-0.39	0.700
District 23	3.54	2.67	31.74	1.33	0.194
District 29	3.03	4.40	45.65	0.69	0.495
District 31	-3.78	2.28	31.92	-1.66	0.107
District 35	-3.13	2.09	32.21	-1.50	0.143
District 51	6.63	2.97	36.34	2.23	0.032
District 57	4.81	3.09	32.28	1.56	0.130
Treatment*Black	1.86	2.39	4,971.38	0.78	0.436
Treatment*White	3.34	2.44	4,972.19	1.37	0.171
Treatment* Latino/a/e	3.70	2.92	4,977.39	1.27	0.205
Treatment*Grade 4	0.97	8.89	4,938.24	0.11	0.914
Treatment*Grade 5	4.30	1.96	482.81	2.20	0.029
Treatment*Grade 6	4.14	1.87	610.00	2.22	0.027
Treatment*Grade 7	3.10	1.76	904.51	1.76	0.078
Treatment*English Learner	-2.13	3.41	4,966.07	-0.62	0.533
Treatment*Economically Disadvantaged	1.28	1.33	4,977.18	0.96	0.338
Treatment*Special Education	-0.89	1.35	4,959.03	-0.66	0.512

**Table D2**

Comparison Group 2: Analysis of adequate usage on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	729.73	0.63	36.88	1,162.08	0.000
Usage (90 Lessons)	6.12	0.65	4,719.71	9.45	0.000
2022-2023 LEAP Math Score	0.79	0.01	4,985.50	81.02	0.000
Black or African American	-2.74	1.11	4,981.02	-2.46	0.014
White	-0.25	1.13	4,975.05	-0.22	0.828
Latino/a/e	0.63	1.37	4,984.12	0.46	0.648
English Language Learner Status	-2.17	1.70	4,971.79	-1.28	0.201
Economically Disadvantaged Status	-3.37	0.66	4,987.73	-5.07	0.000
Special Education Status	-4.51	0.69	4,963.57	-6.57	0.000
Black/African American Enrollment Rate	-0.07	3.91	38.72	-0.02	0.985
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.14	0.06	36.71	2.44	0.020
District 8	-2.86	3.45	29.97	-0.83	0.415
District 23	3.49	2.74	31.37	1.27	0.212
District 29	4.77	4.50	44.16	1.06	0.295
District 31	-1.16	2.73	30.72	-0.43	0.673
District 35	3.55	2.75	30.64	1.29	0.206
District 48	6.07	2.55	32.11	2.38	0.023
District 51	6.62	3.02	34.79	2.19	0.035
District 57	4.67	3.17	31.68	1.47	0.150
Grade 4	20.18	4.25	4,944.91	4.75	0.000
Grade 5	-0.23	1.01	804.26	-0.23	0.818
Grade 6	-1.39	0.94	876.67	-1.47	0.141
Grade 7	0.01	0.89	1,130.53	0.01	0.993

**Table D3**

Comparison Group 2: Analysis of usage level on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	726.54	0.72	69.73	1,015.19	0.000
Low Zearn Usage	2.44	0.66	4,904.48	3.70	0.000
Medium Zearn Usage	7.01	0.71	4,333.37	9.80	0.000
High Zearn Usage	10.14	0.79	4,062.08	12.88	0.000
2022-2023 LEAP Math Score	0.78	0.01	4,986.46	79.43	0.000
Black or African American	-3.04	1.10	4,981.13	-2.75	0.006
White	-0.51	1.12	4,975.22	-0.46	0.647
Latino/a/e	0.35	1.36	4,983.21	0.26	0.794
English Learner	-2.50	1.69	4,971.46	-1.48	0.139
Economically Disadvantaged Status	-3.38	0.66	4,986.60	-5.14	0.000
Special Education Status	-4.20	0.68	4,963.63	-6.16	0.000
Black/African American Enrollment Rate	1.96	3.76	38.67	0.52	0.605
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.19	0.06	36.95	3.32	0.002
District 8	-3.85	3.29	29.30	-1.17	0.252
District 23	3.75	2.62	30.74	1.44	0.161
District 29	6.36	4.33	44.44	1.47	0.149
District 31	-2.34	2.61	30.16	-0.90	0.376
District 35	3.69	2.62	30.01	1.41	0.170
District 48	5.15	2.43	31.51	2.12	0.042
District 51	6.76	2.89	34.20	2.34	0.025
District 57	4.28	3.03	31.06	1.41	0.168
Grade 4	18.93	4.21	4,936.85	4.50	0.000
Grade 5	-0.40	0.99	724.24	-0.41	0.683
Grade 6	-1.47	0.93	786.47	-1.58	0.113
Grade 7	-0.04	0.87	1,021.03	-0.05	0.959

**Table D4**

*Comparison Group 2: Analysis of total on-grade completed lessons on spring 2024
LEAP Math overall scale scores*

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	725.29	0.74	76.70	977.02	0.000
Total Completed On-Grade Lessons	0.10	0.01	3,692.33	13.37	0.000
2022-2023 LEAP Math Score	0.77	0.01	4,985.63	78.82	0.000
Black or African American	-2.92	1.11	4,982.45	-2.64	0.008
White	-0.36	1.12	4,976.24	-0.32	0.748
Latino/a/e	0.51	1.36	4,984.94	0.38	0.705
English Learner	-2.57	1.69	4,973.30	-1.52	0.128
Economically Disadvantaged Status	-3.35	0.66	4,988.35	-5.08	0.000
Special Education Status	-4.22	0.68	4,963.51	-6.19	0.000
Black/African American Enrollment Rate	2.03	3.80	39.04	0.53	0.596
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.20	0.06	37.53	3.46	0.001
District 8	-3.58	3.33	29.66	-1.07	0.292
District 23	4.06	2.65	31.15	1.53	0.135
District 29	6.24	4.37	44.65	1.43	0.161
District 31	-2.46	2.63	30.33	-0.94	0.357
District 35	4.12	2.65	30.38	1.55	0.131
District 48	5.38	2.46	31.87	2.18	0.036
District 51	6.76	2.92	34.57	2.32	0.027
District 57	4.54	3.06	31.40	1.48	0.148
Grade 4	18.71	4.21	4,943.17	4.44	0.000
Grade 5	-0.82	0.99	786.51	-0.83	0.409
Grade 6	-1.77	0.93	846.02	-1.90	0.058
Grade 7	-0.20	0.87	1,066.42	-0.23	0.817

**Table D5**

Comparison Group 2: Analysis of average lessons completed per week on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	724.29	0.80	74.71	910.69	0.000
Average Lessons Completed Per Week	3.33	0.22	4,459.69	15.16	0.000
2022-2023 LEAP Math Score	0.76	0.01	4,945.76	75.97	0.000
Black or African American	-2.07	1.12	4,935.36	-1.86	0.063
White	0.28	1.14	4,929.01	0.25	0.805
Latino/a/e	1.01	1.37	4,938.65	0.74	0.460
English Learner	-2.26	1.68	4,927.70	-1.35	0.178
Economically Disadvantaged Status	-3.32	0.66	4,942.99	-5.05	0.000
Special Education Status	-4.42	0.68	4,920.20	-6.51	0.000
Black/African American Enrollment Rate	2.14	4.15	40.06	0.52	0.609
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.19	0.06	38.27	3.12	0.003
District 8	-4.19	3.69	31.81	-1.14	0.265
District 23	4.88	2.93	33.32	1.67	0.105
District 29	7.82	4.76	44.69	1.64	0.107
District 31	-0.73	2.90	31.77	-0.25	0.802
District 35	5.16	2.94	32.58	1.76	0.088
District 48	6.81	2.71	33.83	2.51	0.017
District 51	7.87	3.21	36.48	2.45	0.019
District 57	5.94	3.38	33.48	1.76	0.088
Grade 4	19.43	4.19	4,921.10	4.64	0.000
Grade 5	-0.61	1.01	1,039.86	-0.60	0.547
Grade 6	-1.37	0.94	1,107.32	-1.45	0.148
Grade 7	0.10	0.88	1,420.31	0.12	0.907



APPENDIX E: Comparison Group 3 Analytic Models

Table E1

Comparison Group 3: Impact analysis of Zearn Supplemental on spring 2024 LEAP Math overall scale scores: Full model

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	731.23	1.23	36.86	596.48	0.000
Treatment	2.59	2.24	37.45	1.16	0.253
Black or African American	-6.32	1.56	4,532.58	-4.05	0.000
White	-2.28	1.48	4,505.28	-1.54	0.123
Latino/a/e	-4.99	1.96	4,511.89	-2.55	0.011
Grade 4	30.38	4.37	3,002.08	6.96	0.000
Grade 5	5.23	1.92	244.74	2.72	0.007
Grade 6	6.31	1.91	235.76	3.30	0.001
Grade 7	7.83	1.75	501.33	4.47	0.000
English Learner	2.00	2.36	4,509.06	0.84	0.398
Economically Disadvantaged Status	-1.64	0.82	4,530.59	-2.00	0.045
Special Education Status	-5.28	0.93	4,512.67	-5.67	0.000
2022-2023 LEAP Math Score	0.80	0.01	4,532.99	85.74	0.000
Black/African American Enrollment Rate	-2.43	3.58	40.89	-0.68	0.502
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.01	0.05	39.18	0.14	0.885
District 8	-2.22	2.79	38.30	-0.79	0.432
District 29	0.15	3.26	41.02	0.05	0.964
District 31	-1.34	2.32	35.40	-0.58	0.566
District 35	-2.78	2.12	34.96	-1.31	0.198
Treatment*Black	5.62	2.40	4,527.55	2.34	0.019
Treatment*White	4.72	2.36	4,520.30	2.00	0.045
Treatment* Latino/a/e	8.87	2.90	4,509.07	3.06	0.002
Treatment*Grade 4	-17.55	7.20	4,212.56	-2.44	0.015
Treatment*Grade 5	-2.43	2.27	297.13	-1.07	0.286
Treatment*Grade 6	-3.54	2.25	301.88	-1.57	0.117
Treatment*Grade 7	-5.15	2.11	628.37	-2.44	0.015
Treatment*English Learner	-6.68	3.59	4,508.62	-1.86	0.062
Treatment*Economically Disadvantaged	-1.56	1.24	4,516.00	-1.26	0.209
Treatment*Special Education	0.96	1.36	4,516.59	0.70	0.482

**Table E2**

Comparison Group 3: Analysis of adequate usage on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	730.94	0.62	43.64	1,170.72	0.000
Usage (90 Lessons)	5.77	0.64	4,064.47	8.97	0.000
2022-2023 LEAP Math Score	0.78	0.01	4,541.98	81.27	0.000
Black or African American	-4.01	1.18	4,536.47	-3.41	0.001
White	-0.40	1.14	4,519.91	-0.35	0.724
Latino/a/e	-1.01	1.42	4,517.80	-0.72	0.474
English Learner	-1.49	1.76	4,515.52	-0.85	0.396
Economically Disadvantaged Status	-2.15	0.62	4,541.12	-3.47	0.001
Special Education Status	-4.84	0.69	4,516.66	-6.99	0.000
Black/African American Enrollment Rate	-1.47	3.63	43.58	-0.40	0.688
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.05	0.05	42.39	1.04	0.306
District 8	-1.71	2.80	39.03	-0.61	0.545
District 29	0.96	3.24	40.29	0.30	0.768
District 31	-1.62	2.58	38.28	-0.63	0.533
District 35	0.08	2.53	36.65	0.03	0.976
District 48	2.38	2.22	36.93	1.07	0.291
Grade 4	21.44	3.38	4,453.54	6.35	0.000
Grade 5	2.02	1.06	774.44	1.91	0.057
Grade 6	2.68	1.05	771.31	2.56	0.011
Grade 7	4.18	0.99	1,210.02	4.24	0.000

**Table E3**

Comparison Group 3: Analysis of usage level on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	729.06	0.76	83.78	961.94	0.000
Low Zearn Usage	1.46	0.67	4,515.86	2.17	0.030
Medium Zearn Usage	3.96	0.72	4,344.46	5.49	0.000
High Zearn Usage	8.16	0.80	3,657.63	10.17	0.000
2022-2023 LEAP Math Score	0.77	0.01	4,538.88	79.31	0.000
Black or African American	-4.19	1.17	4,533.26	-3.57	0.000
White	-0.49	1.14	4,517.67	-0.43	0.664
Latino/a/e	-1.20	1.41	4,515.20	-0.85	0.397
English Learner	-1.31	1.75	4,513.04	-0.75	0.456
Economically Disadvantaged Status	-2.04	0.62	4,539.27	-3.31	0.001
Special Education Status	-4.70	0.69	4,514.52	-6.81	0.000
Black/African American Enrollment Rate	-0.65	3.72	44.04	-0.17	0.863
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.07	0.05	43.59	1.47	0.149
District 8	-2.04	2.87	39.52	-0.71	0.482
District 29	0.82	3.32	40.72	0.25	0.806
District 31	-2.34	2.65	38.96	-0.88	0.383
District 35	0.04	2.60	37.25	0.01	0.989
District 48	1.93	2.28	37.54	0.85	0.403
Grade 4	20.57	3.37	4,461.82	6.10	0.000
Grade 5	1.37	1.07	864.37	1.28	0.202
Grade 6	1.87	1.06	875.01	1.76	0.079
Grade 7	3.55	1.00	1,323.79	3.57	0.000

**Table E4**

*Comparison Group 3: Analysis of total on-grade completed lessons on spring 2024
LEAP Math overall scale scores*

Coefficient	Estimate	SE	df	t value	p value
(Intercept)	727.74	0.78	88.09	933.08	0.000
Total Completed On-Grade Lessons	0.08	0.01	3,431.23	10.21	0.000
2022-2023 LEAP Math Score	0.76	0.01	4,539.10	78.54	0.000
Black or African American	-4.16	1.17	4,535.15	-3.54	0.000
White	-0.46	1.14	4,519.20	-0.41	0.683
Latino/a/e	-1.10	1.41	4,517.03	-0.78	0.437
English Learner	-1.28	1.75	4,514.91	-0.73	0.464
Economically Disadvantaged Status	-2.05	0.62	4,541.48	-3.32	0.001
Special Education Status	-4.72	0.69	4,515.88	-6.84	0.000
Black/African American Enrollment Rate	-0.63	3.76	44.02	-0.17	0.869
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.08	0.05	43.86	1.65	0.107
District 8	-2.34	2.90	39.58	-0.81	0.424
District 29	0.49	3.35	40.79	0.15	0.885
District 31	-2.33	2.67	39.11	-0.87	0.389
District 35	0.11	2.63	37.35	0.04	0.967
District 48	1.84	2.31	37.66	0.80	0.431
Grade 4	20.24	3.38	4,466.84	6.00	0.000
Grade 5	0.97	1.08	909.93	0.90	0.368
Grade 6	1.43	1.07	920.14	1.33	0.183
Grade 7	3.08	1.00	1,353.08	3.09	0.002

**Table E5**

Comparison Group 3: Analysis of average lessons completed per week on spring 2024 LEAP Math overall scale scores

Coefficient	Estimate	SE	df	<i>t</i> value	<i>p</i> value
(Intercept)	726.88	0.80	88.16	913.57	0.000
Average Lessons Completed Per Week	2.61	0.22	4,002.33	11.84	0.000
2022-2023 LEAP Math Score	0.76	0.01	4,524.98	77.31	0.000
Black or African American	-3.99	1.17	4,516.53	-3.42	0.001
White	-0.38	1.13	4,501.31	-0.34	0.736
Latino/a/e	-1.14	1.41	4,498.90	-0.81	0.419
English Learner	-1.80	1.75	4,498.09	-1.03	0.302
Economically Disadvantaged Status	-2.02	0.61	4,524.91	-3.30	0.001
Special Education Status	-4.85	0.69	4,498.29	-7.06	0.000
Black/African American Enrollment Rate	-0.54	3.85	43.77	-0.14	0.890
% of Students 2022-2023 LEAP Mastery+ Grade 3-8	0.08	0.05	43.21	1.50	0.142
District 8	-2.46	2.98	39.56	-0.83	0.414
District 29	1.62	3.44	40.79	0.47	0.640
District 31	-1.12	2.73	38.32	-0.41	0.683
District 35	0.82	2.70	37.52	0.30	0.763
District 48	2.81	2.36	37.71	1.19	0.241
Grade 4	21.04	3.35	4,457.58	6.28	0.000
Grade 5	1.31	1.08	950.46	1.22	0.224
Grade 6	2.00	1.06	939.79	1.88	0.060
Grade 7	3.36	1.00	1,414.50	3.37	0.001

